

Multi-Modal Protein Function Prediction using a Joint Embedding Space from Two Graph Neural Networks

by

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ABSTRACT

In bioinformatics and proteomics, determining protein functions experimentally is expensive and slow. There's a growing need for precise and quick computational prediction methods, filling the gap between sequence discovery and functional understanding. Over recent years there has been an influx of deep-learning protein folding algorithms used for predicting function by transfer learning. Protein function is only partially captured by each of a large number of modalities including structure, however, in isolation they only give us a partial understanding of function. Uniting these is an important step to understanding function more holistically. We present a multi-modal framework using two graph neural networks to infer a joint embedding space that captures many properties of a protein including structure, disease associations, drug interactions, protein interactions, biological processes and more. We evaluate the embedding space on downstream prediction tasks including enzyme commission (EC) numbers and gene ontology (GO) terms. Experimental results on protein function prediction, as well as a qualitative visual analysis of the protein embedding space show that our framework is able to successfully capture both structure and biomedical context of proteins, and outperforms structure-only based encoders.

Thesis supervisor: Manolis Kellis

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Chapter 1

Introduction

Proteins are vital elements of the cell and are involved in a wide array of applications, from medical treatments to biological engineering to materials science . Recent years have seen an explosion in the discovery of new protein sequences, in part due to the advance of affordable sequencing technologies. However, the process of assigning functions to these new protein sequences is still expensive and slow. The function of more than 30% of proteins remains unknown [1]. There’s a growing need for precise and quick computational methods to predict protein functions, filling the gap between sequence discovery and functional understanding.

Because many functions are in part dictated by their respective structures, there exist numerous deep learning approaches to predict these protein functions from structure [2]. Given the significant challenge in determining protein structures and due to the nearly 200-fold difference in the number of experimentally determined protein sequences compared to structures [3], most data-driven methods have favored a sequence-to-function approach. The landscape shifted with the introduction of highly-accurate, deep-learning protein folding algorithms such as AlphaFold [4] and ESMFold [5], along with other notable advancements, providing the necessary data to explore a structure-to-function approach.

Since then many methods have been developed to predict function from structure. However, while structure is a critical factor in determining a protein’s function, it’s not the sole

factor. Function is captured in part by gene expression, in part by structure and in part by disease associations, drug interactions, gene ontologies, etc. Each of these modalities are highly informative of function, but not in isolation. A multi-modal model that considers various protein properties, offers a more comprehensive understanding of protein function. The field of multi-modal models is relatively new and inherently complex. One of the main challenges lies in integrating various types of data – such as structural information, genetic context, and biochemical properties – in a way that is both computationally feasible and biologically meaningful. Such integration requires a deep understanding of the biological relevance of each data type and ensuring the model can capture such relevance.

Motivated by this idea, my thesis presents a framework using two graph neural networks to develop a joint protein embedding space that captures structure and biomedical information, in order to successfully predict protein function.

1.1 Related Work

A topic of growing interest in deep learning are foundation models. Foundation models, a term coined by researchers at Stanford University [6], refer to a class of deep learning models pretrained on a broad range of data at scale, that serve as a starting point for transfer learning. These models, such as BERT [7] for language and GPT [8] for generative tasks, have significantly impacted various fields, demonstrating remarkable flexibility and efficiency. Just as language models learn the syntax of human language, biological foundation models aim to learn the language of proteins – patterns in sequences and structures that dictate their properties. By pretraining on large datasets of DNA sequences [9], single-cell transcriptomics data [10], single-cell RNA sequencing data [11][12], or proteins [13][14], these models can capture the relationships that underpin interactions and functions. For protein representation learning, many models have been presented, each targeting different features of a protein. Sequence-based models such as UniRep[15] and SeqVec[16], use LSTM archi-

structures to capture long-range dependencies in protein sequences. Structure-based models such as ProNet[14] and GearNet[13], aim to encode spatial information about the protein structures.

Another family of models, supervised algorithms, have emphasized the importance of protein structure in function determination. Architectures such as 3D CNNs [17] and GNNs [2][18] are adopted to capture the spatial complexities inherent in protein structure. For example, IEConv [19] introduced a graph convolution layer sensitive to both intrinsic and extrinsic node distances, aligning with protein structure modeling biases. These structural models are considered more effective for protein representation compared to sequential methods or multiple sequence alignments (MSAs).

Knowledge graphs (KG) are a way of representing multi-modal data – a capability particularly useful in bioinformatics, given the variety of data types such as text, gene expression, sequences, images, and more. Several works have been devoted to constructing knowledge graphs for different purposes, including drug discovery, disease prediction, and molecular biology. A group at Harvard released a knowledge graph for precision medicine that describes almost 20,000 different diseases [20], while other groups used knowledge graphs for disease prediction tasks [21]. In the specific context of proteins, another group introduced STRING, a database and KG of known and predicted protein-protein interactions [22]. While promising, the scope of these efforts has primarily been limited to protein-protein interactions and doesn't encompass the entire gamut of protein-related information. Parallel to this, there has been work done applying GNNs to the biomedical domain. TxGNN was recently trained on a knowledge graph for drug repurposing, presenting the use of enhanced embeddings to improve prediction capabilities for poorly characterized diseases [23]. The combination of knowledge graphs with deep learning such as LLMs and GNNs allows us to benefit from the strengths of both [24].

1.2 Contributions

My thesis proposes a joint embedding space inferred from two graph neural networks (GNNs) that encodes multiple protein properties – including associated drugs, diseases, functions, pathways, structure, sequence, etc. – which can then be used for downstream tasks such as function prediction. The model is composed of two modules. First, ProtGNN, is a self-supervised, foundation model that learns protein representations based off of a biomedical knowledge graph. The knowledge graph contains various node types including, but not limited to, proteins, drugs, diseases, molecular functions, pathways and biological processes. The idea is that by training a GNN (ProtGNN) on this knowledge graph, it learns node representations that reflect the complex interplay of relationships and attributes within the biomedical domain.

The second component leverages these learned representations from ProtGNN, specifically the protein representations, as targets for a subsequent GNN with protein structure graphs as input. This second GNN, GearNet-ProtGNN, is able to map from protein structure to a protein embedding space that captures biomedical knowledge. In doing so, protein function can be effectively predicted via transfer learning with GearNet-ProtGNN by integrating various modalities into the protein representations.

Another novel aspect of this framework is that the joint embedding space is created without the need for an alignment module and parallel training of two separate GNNs. There are many use cases of alignment modules in the context of GNNs including graph matching [25], node alignment [26] and domain adaptation where it is necessary to combine information from multiple graphs. Alignment modules compute the embedding space for two models simultaneously on a loss function that minimizes the difference between the representations of two of the same entities. In contrast, the method I adopt trains two GNNs sequentially by using the embedding space generated by the first as a target for the second, and thus streamlines the learning process.

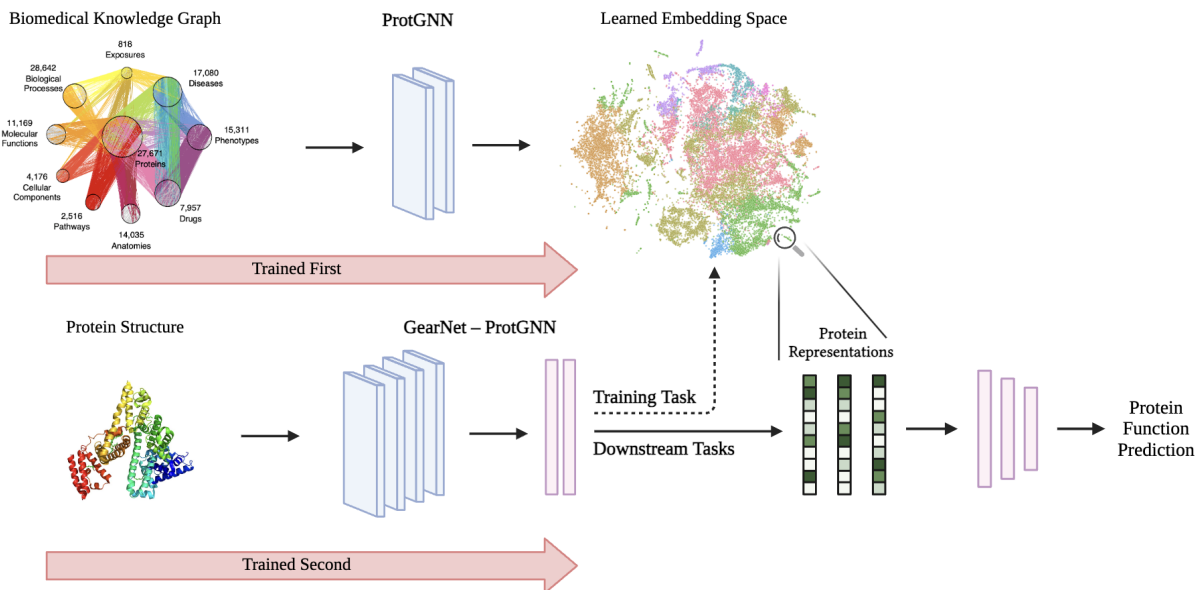


Figure 1.1: Framework for a double GNN system with a joint embedding space used to learn protein representations combining structural and biomedical information.

1.3 Thesis outline

In Chapter 2, I present a GNN model that learns protein representations which encode biomedical contextual information – **ProtGNN**. The protein representations can be used for other tasks that would benefit from understanding the biomedical context around proteins. This model serves as the first module of our multi-hierarchical GNN framework. I first describe prior work pertaining to the use of knowledge graphs in the biomedical space as well as protein representation methods. Next I delve into the architecture of ProtGNN and training strategies implemented, including testing whether protein sequence information not present in the original KG would boost the model performance. I assess the model’s performance qualitatively by visualizing the embedding space, and quantitatively by running a silhouette cluster separation analysis. Finally, I perform an ablation study to understand which node types contribute most to the protein representations.

In Chapter 3, I develop a GNN that predicts ProtGNN protein representations from

protein structure. The idea is that **GearNet-ProtGNN** integrates structural information of proteins with their biomedical context to enhance function prediction capabilities. This approach leverages the strengths of both ProtGNN and GearNet to create a more comprehensive and accurate model. This is the second module of our multi-hierarchical GNN framework. I first describe the current state of the field for protein function prediction from structure and protein structure models. Next, I lay out the model’s architecture and the unique training task. For evaluation, I describe four downstream tasks and analyze the results of how GearNet-ProtGNN performs compared to ProtGNN and GearNet separately. Finally, I analyze the embedding space predicted by GearNet-ProtGNN and see how it differs from target embedding space.

Chapter 2

Learned Protein Representations

2.1 Motivation

Leveraging protein representations encoding biomedical information offers a promising approach in the realm of structure-to-function prediction in proteomics. These representations, originally trained to understand and predict functional relationships between proteins, inherently encode a wealth of contextual information beyond a protein’s function. Crucially, they encode details about associated diseases, relevant drug interactions, and broader biological processes and pathways. This multi-dimensional representation can serve as a proxy for protein function in other models aimed at deciphering the complex relationship between a protein’s structure and its function, leading to a more holistic understanding of protein function and more accurate predictions.

2.2 Prior Work

2.2.1 Biomedical knowledge graph

I use a large-scale biomedical knowledge graph curated by Professor Marinka Zitnik’s group – PrimeKG – that contains 27,671 proteins [20]. PrimeKG compiles data from databases

that cover a broad variety of biomedical information including human disease, drug-protein interactions, genes and proteins with their associated biological processes, functions and cellular component, etc. PrimeKG contains 10 different node types – shown in Figure 2.1 – and 29 different types of undirected edges. Information about associations between proteins and biological processes and molecular functions were retrieved from Entrez Gene [27]. There are 129,375 nodes and 4,050,249 edges in total. For learning protein representations centered around their biological processes and molecular functions, I finetune on the protein-molecular function and protein-biological process edges. There are 139,060 and 289,610 of each edge type respectively.

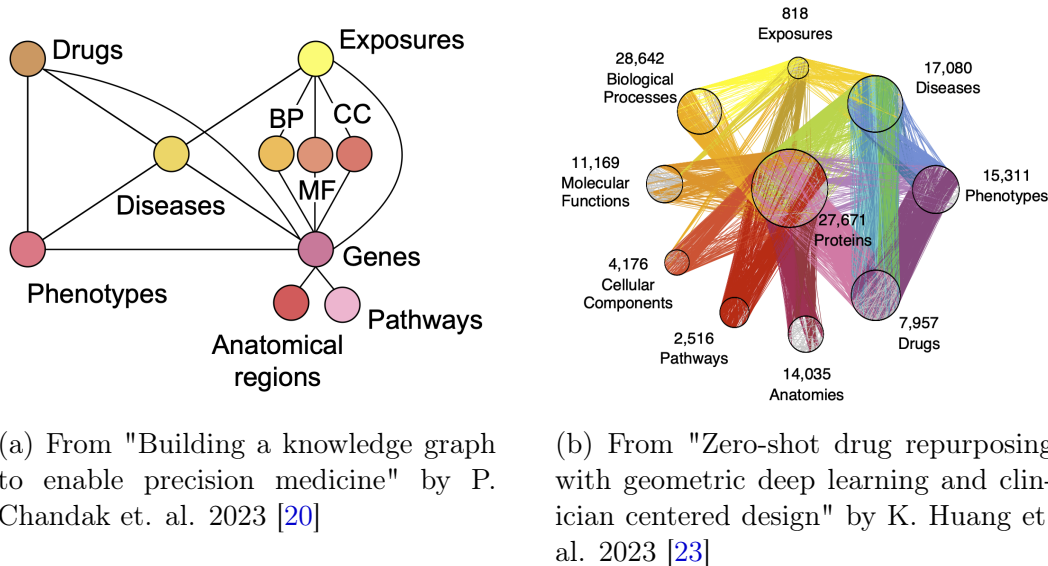


Figure 2.1: PrimeKG contains 10 different node types and provides a comprehensive picture of the biomedical space.

2.2.2 Baseline graph neural network

For my baseline model, I used TxGNN which was applied to PrimeKG for drug repurposing [23]. TxGNN uses a GNN architecture to embed drugs and diseases into a latent space that is optimized to reflect the geometry of the biomedical knowledge graph, PrimeKG. This is the first step of TxGNN – large-scale pretraining on the entirety of the knowledge graph in order to learn meaningful representations of all nodes. The second step was finetuning this

pretrained model to predict drug and disease links using zero-shot learning. TxGNN serves as the backbone to ProtGNN so that it can learn the geometry of PrimeKG and later be finetuned for tasks relating to proteins and their functions.

2.3 Methodology

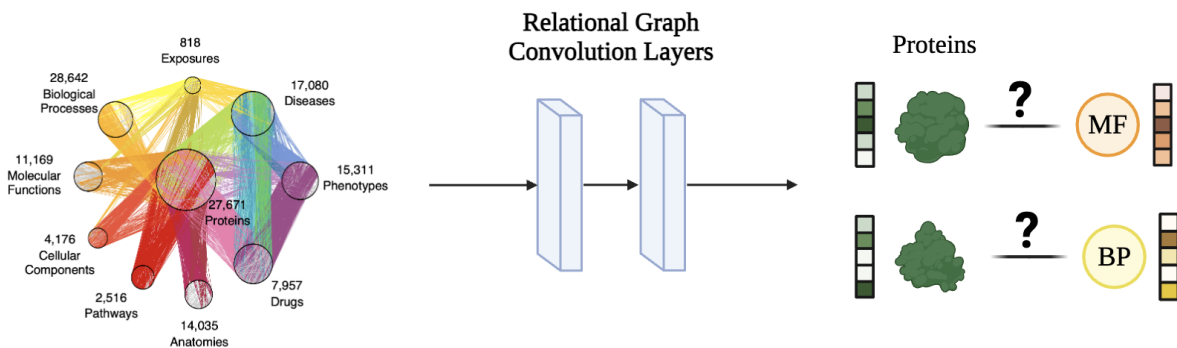


Figure 2.2: Architecture overview of a relational GNN for predicting protein-function and protein-biological process links in a biomedical knowledge graph.

2.3.1 Graph neural network message-passing

During the training phase, ProtGNN uses a standard message-passing algorithm to update and optimize the node embeddings. There is a relation-type specific weight matrix for each of the 29 edge types, denoted as $\mathbf{W}_r$, used to calculate relation-type specific messages. For each node i , incoming messages from neighboring nodes are aggregated for each relation-type r , denoted $\mathcal{N}_r(i)$. This is done by taking the average of these messages:

$$\mathbf{u}_i^{(l)} = \sum_{r \in \mathcal{R}} \frac{1}{|\mathcal{N}_r(i)|} \left(\mathbf{W}_r \sum_{j \in \mathcal{N}_r(i)} \mathbf{h}_j^{(l-1)} \right) \quad (2.1)$$

The new node embedding is then updated by combining the node embedding from the

previous layer and the aggregated messages from all relations:

$$\mathbf{h}_i^{(l)} = \mathbf{h}_i^{(l-1)} + \mathbf{u}_i^{(l)} \quad (2.2)$$

ProtGNN uses 2 R-GCN layers, thus propagating messages between neighboring nodes twice.

2.3.2 Training

I train ProtGNN using a pretraining strategy, during which the model is trained to predict links between nodes of all relation types in PrimeKG using stochastic minibatching. This enforces that ProtGNN learns node representations, for all node types, that best capture the biomedical information in each respective neighborhood.

Next, I finetune ProtGNN with the objective to predict protein-function and protein-biological process links – a standard knowledge graph link-prediction task for GNNs. ProtGNN is finetuned on predicting only these two types of relations. For instance, given protein x and biological process y , what is the probability of a link between this protein-biological process pair? DistMult is used to calculate the probability of that link, where $\sigma(\cdot)$ is the sigmoid function, $\mathbf{w}_r$ is the learned relation-specific weight vector, and $\mathbf{h}_x$ and $\mathbf{h}_y$ are the learned node embeddings.

$$p_{x,y} = \sigma(\mathbf{h}_x \cdot \mathbf{w}_r \cdot \mathbf{h}_y) \quad (2.3)$$

The loss for the finetuning objective is binary cross entropy loss (BCE) between the predicted probability of linkage and the true link (whether or not a link actually exists between protein x and biological process y in the knowledge graph). Formally this is denoted as:

$$\mathcal{L}_{x,y} = \text{BCE}(p_{x,y}, \mathbb{I}(e_{x,y} \in \mathcal{E})) \quad (2.4)$$

Model hyperparameters were optimized by a hyperparameter sweep conducted in Weights &

Biases. During hyperparameter tuning of pretraining hyperparameters, the finetuning hyperparameters were fixed, and vice versa for tuning finetuning hyperparameters. Optimized hyperparameters can be found in Appendix [B.1](#).

2.3.3 Labelling proteins with gene ontology (GO) annotations

Gene Ontology (GO) annotations represent a structured, hierarchical system that categorizes genes across three ontologies: biological processes, molecular functions, and cellular components [28]. In "biological processes", GO annotations describe large processes/programs accomplished by one or more molecular activities, such as cell death or immune responses. "Molecular functions" refer to the elemental activities of a gene at the molecular level, like transporter activity or binding. Lastly, "cellular components" denote the locations relative to cellular structures where a protein performs its function, such as in the nucleus or membrane. The hierarchical nature of GO allows for a multi-level representation, ranging from very general, broad terms to more specific and granular terms, enabling a nuanced and comprehensive understanding of gene functions and their roles in the cell.

Given this hierarchical structure and the fact that every protein has multiple annotations, each protein needed to be labelled with the corresponding top-level biological process. The following steps were used to label each protein with a biological process:

1. Create a mapping between all biological process annotations to their corresponding top-level annotation.
2. For each protein's list of annotations, using the mapping, convert all of the protein's annotations to the top-level annotation.
3. Return the most common top-level annotation for each protein.

An toy example is shown in Figure [2.3](#). Protein *X* has 4 GO annotations: 1, 5, 6, and 9. These GO terms are highlighted in red in the GO hierarchical tree below and in the

corresponding GO mapping. Protein X 's GO annotations are mapped to their respective top-level terms which in this example are 1, 1, 2, and 1. Finally, protein X is labelled with the most common top-level term, GO 1.

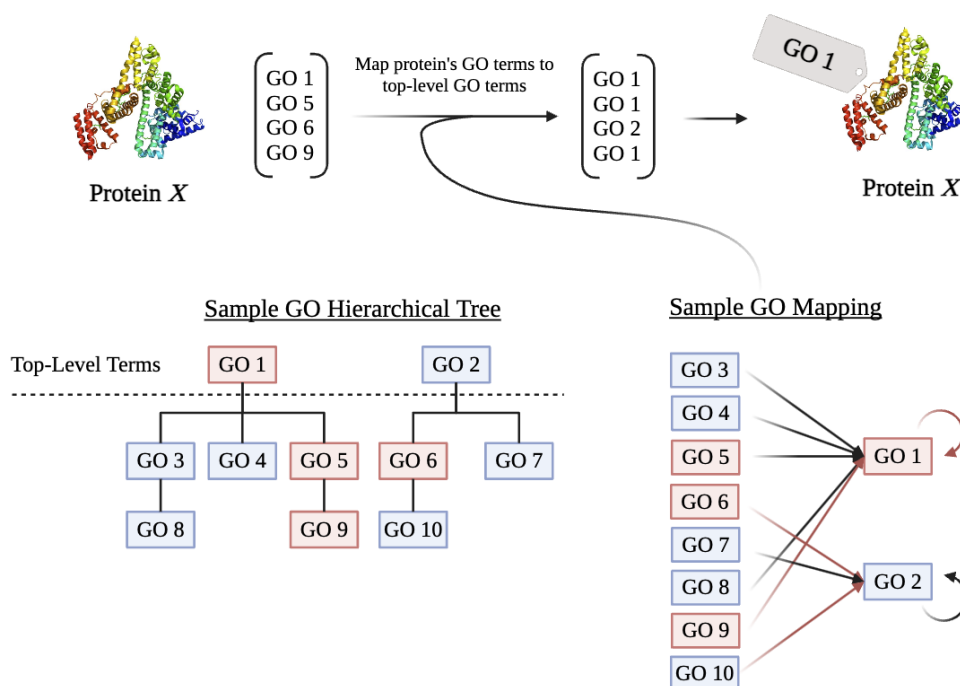


Figure 2.3: Example of GO term labelling of a protein.

The same procedure was repeated for molecular function annotations. Sometimes, the top-level was too broad and non-informative (i.e. "regulation of biological process"), so I also looked at the secondary-level annotations. The difference here was for Step 1, I made a mapping between all annotations and their corresponding secondary-level annotation. In the end, every protein has a top-level and secondary-level biological process label, and a top-level and secondary-level molecular function label.

2.3.4 Silhouette clustering analysis

Silhouette coefficients are a classic metric for determining the goodness of a clustering technique, where values range from -1 to $+1$. It shows how well each object lies within its cluster, which is a reflection of both the compactness and the separation of the clusters. The

silhouette score for each sample is calculated using the following formula:

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)} \quad (2.5)$$

- s_i = silhouette score for a single sample.
- a = mean intra-cluster distance, the average distance between the sample and all other points in the *same* cluster. (compactness)
- b = mean nearest-cluster distance, the average distance between the sample and all points in the *nearest* cluster that the sample is not a part of. (separation)

A high silhouette score close to +1 indicates a sample is well-matched to its own cluster and clusters are well separated. A score close to -1 indicates a sample is poorly matched to its own cluster. A score near 0 indicates overlapping clusters. The silhouette coefficient for the entire embedding space is defined as the max of the mean silhouette scores for each cluster.

$$SC = \max_k \left(\frac{1}{|C_k|} \sum_{i \in C_k} s_i \right) \quad (2.6)$$

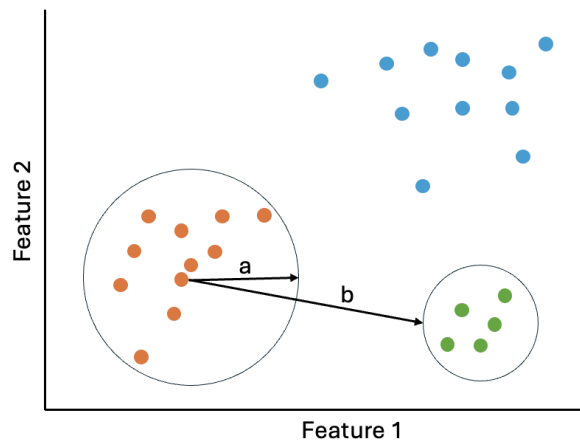


Figure 2.4: Visual depiction of the silhouette score analysis.

In my analysis, the goodness of the protein embedding space can be framed as a goodness of clustering problem: how well does the learned protein embedding space cluster proteins?

Therefore, to quantitatively measure how well the protein embedding space is able to separate proteins based on different labels, I calculated the silhouette coefficient of the protein representation with respect to the relevant protein labels (molecular function or biological process classification).

2.3.5 Testing model performance for different finetuning tasks

One experiment I ran was exploring different finetuning tasks that are central to understanding protein functions. Specifically, I focused on link-prediction tasks between protein nodes and their associated molecular functions (MF) and biological processes (BP). The first task was to predict protein–MF links. The second task was to predict protein–BP links. The final finetuning task predicted both protein–MF and protein–BP links. This multifaceted approach allowed us to determine the model’s ability to predict inter-node relationships. I found that there was no significant difference in model performance between the 3 tasks (see Appendix C for full results) and the visualized protein embedding spaces clustered similarly between all models. Therefore, the model finetuned on both protein–MF and protein–BP links is used for all future analyses.

2.4 Evaluation

2.4.1 Visualizing the protein embedding space

In the fast-evolving world of deep learning, the analysis of model embedding spaces has emerged as an interesting area of study. These embedding spaces are important to understanding how complex models like GNNs prioritize and process the intricate relationships and structures inherent in graph data.

As a verification of concept I looked at the embedding space of all nodes in PrimeKG after training. I used t-distributed Stochastic Neighbor Embedding (t-SNE) algorithm for

dimensionality reduction (from 1024 to 2 dimensions) and colored all points by their respective node type. As seen in Figure 2.5, the node embeddings cluster well by node type. This is non-trivial because the node type is never directly encoded in PrimeKG, only indirectly through edge-type weight matrices.

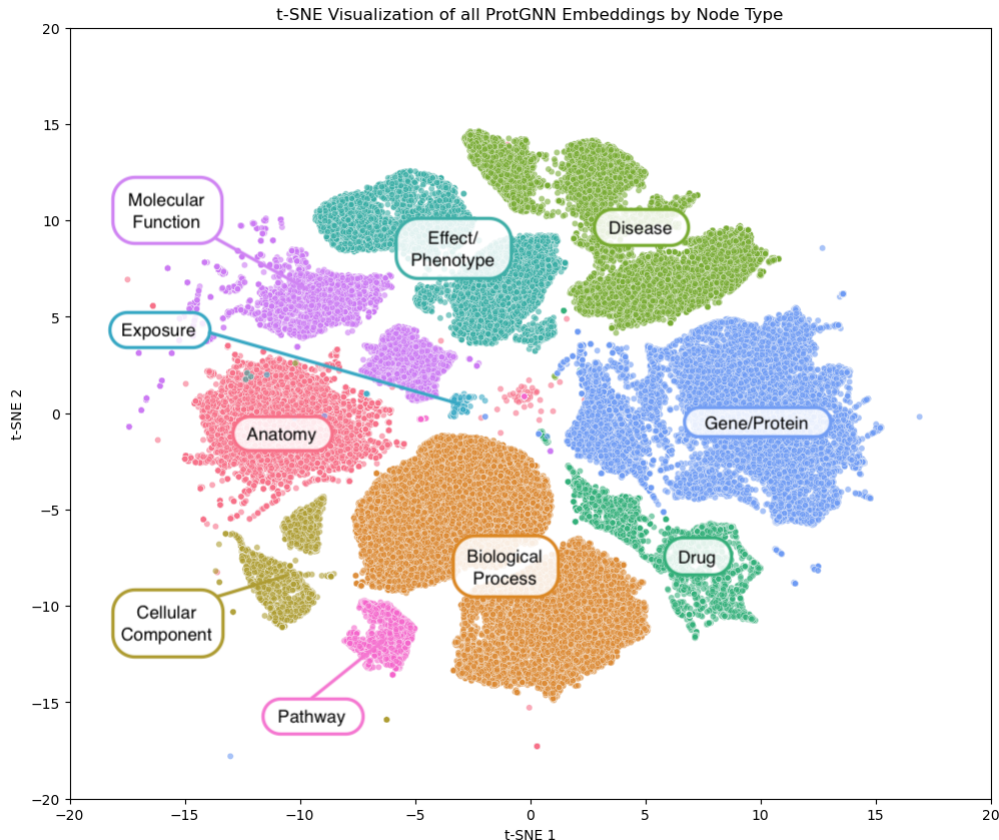


Figure 2.5: Visualization of learned embedding space for all nodes in PrimeKG, colored by node type.

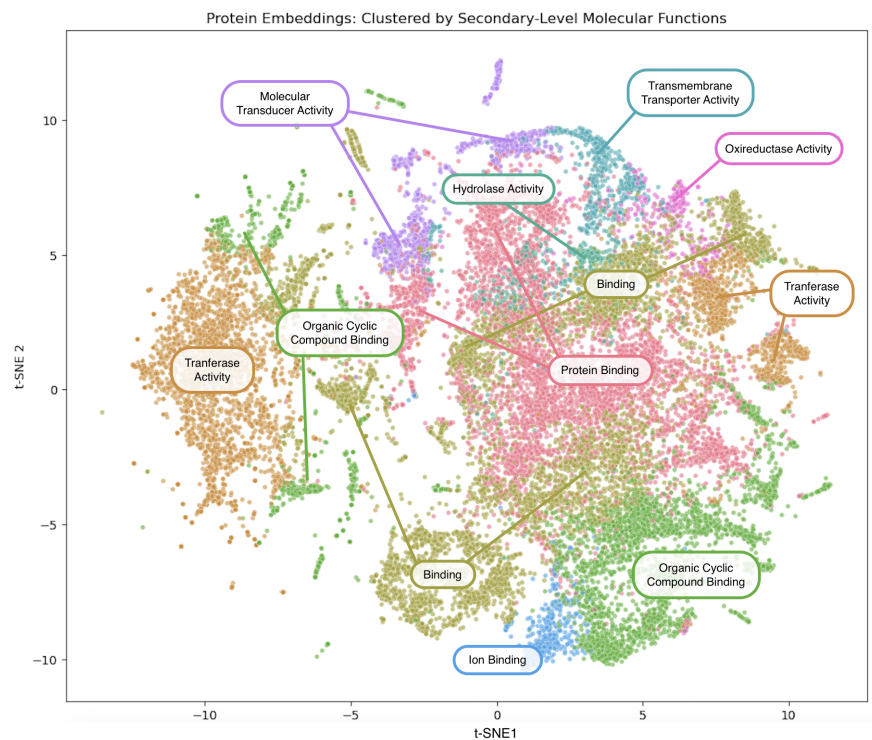
Next, I focus in on the protein embedding space. Specifically, I was interested in visualizing how the protein embedding space clustered by the most enriched GO annotation for each region of the embedding space. Molecular function and biological processes labels are used, as those were the relationships the model was finetuned on. The learned protein embedding space was first clustered using k-means clustering ($k=40$). To get the label for each cluster, the corresponding labels for all proteins in a given cluster (as determined in Section 2.3.3) were aggregated and the most enriched protein label was used as the cluster label. Finally,

to visualize the embedding space, I used t-SNE for dimensionality reduction. In support of the qualitative visual analysis of the latent space, I calculated the silhouette coefficient for all samples with respect to the molecular function or biological process labels.

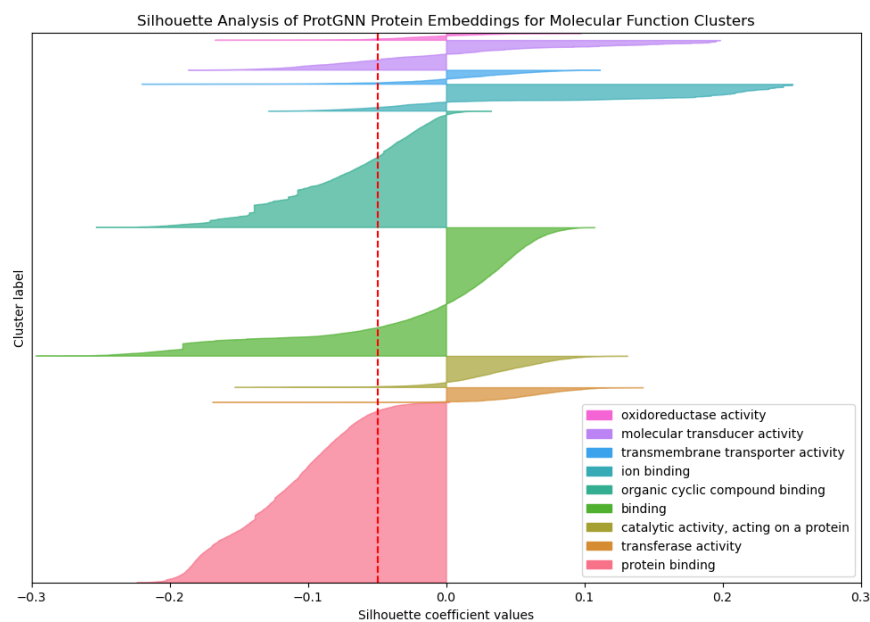
The t-SNE plots of the protein embedding space from ProtGNN show that protein representations successfully encode molecular function and biological process GO annotations. Looking at the protein embedding space colored by molecular function annotations, Figure 2.6 illustrates the efficacy of the model in capturing proteins’ molecular function labels because the proteins cluster well by their labels. Smaller functional groups, defined by fewer associated proteins, tend to form compact clusters. In contrast, larger functional groups, which encompass a greater number of proteins, tend to map to multiple regions across the embedding space. The silhouette analysis presented in Figure 2.6b supports these observations. Larger functional groups such as protein binding, binding, and organic cyclic compound binding – noticeable by their extended distribution along the y-axis of Figure 2.6b – have a significant proportion of proteins with negative silhouette coefficients. This suggests that these groups are overlapping or have less well-defined group boundaries compared to smaller groups. Another observation is that while proteins with the same functional annotation map to the same region in the embedding space (there isn’t a random assortment of colors and therefore functions), there lacks separation between the functional clusters. This can be explained by the fact that the protein representations encode lots of different biomedical information about each protein, ProtGNN isn’t a model trained to only encode the protein-molecular function links in PrimeKG. Therefore it isn’t expected that we can find separable function clusters.

Looking now at the protein embedding space colored by the biological processes in Figure 2.7, a similar conclusion can be made. Cellular metabolic process is the largest group with about half of the proteins labelled with this secondary-level biological process term. Visually, we see a large cellular metabolic cluster on the left side of the embedding space, however there is also a smaller cluster on the right side and in the bottom center of the embedding

space. Smaller process groups such as cell death, immune response, detection of stimulus and cellular development process cluster tightly within themselves and have a high proportion of samples with positive silhouette coefficients. Quantitatively, the protein representations cluster marginally better by biological process compared to molecular function (silhouette coefficient = -0.029 for BP compared to -0.049 for MF, Tables [C.4](#) and [C.3](#)).

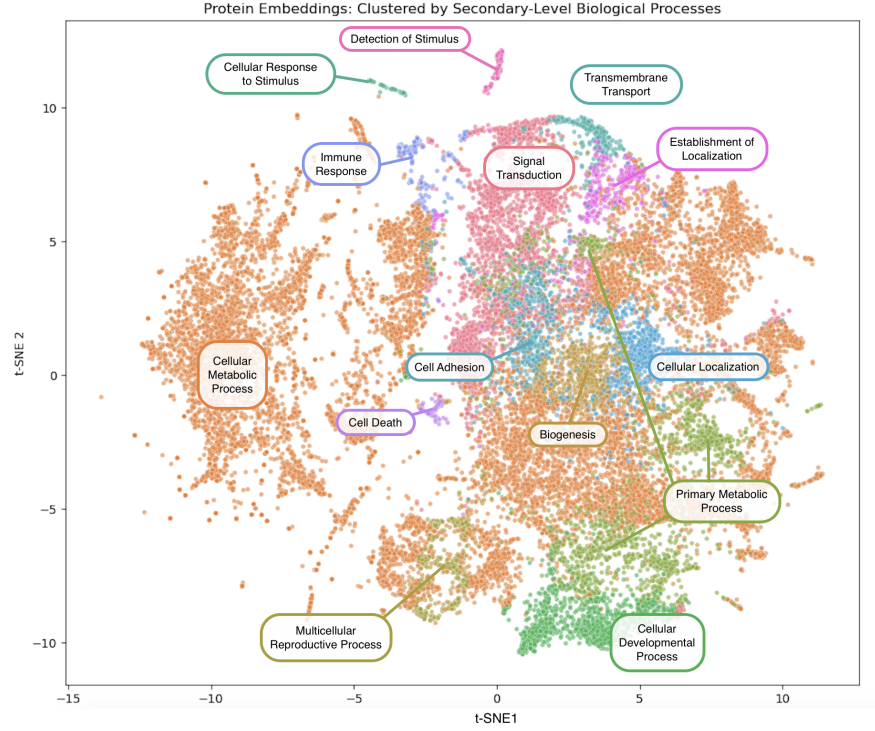


(a) Learned protein embedding space clustered by secondary-level molecular functions.

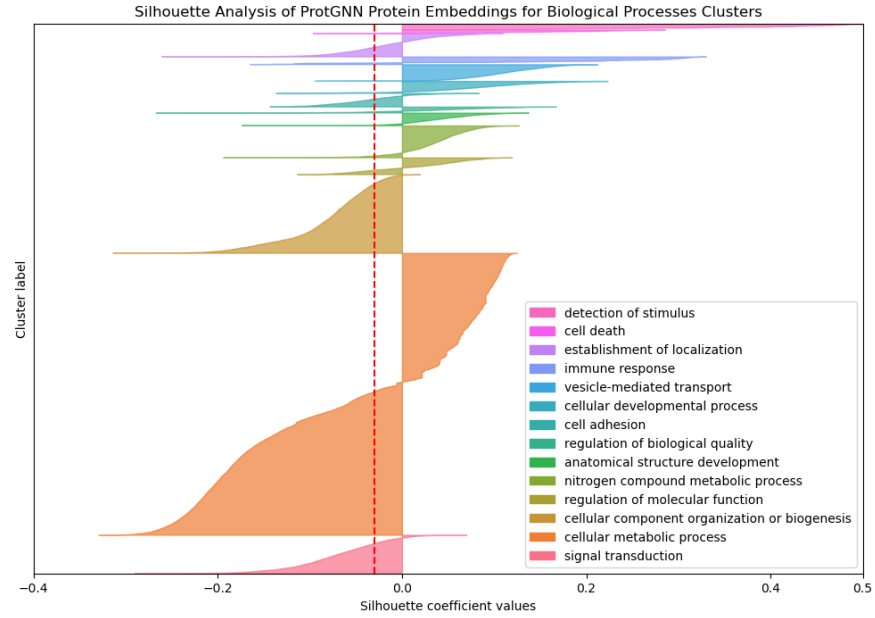


(b) Silhouette scores by molecular function.

Figure 2.6: Visualization of learned protein embedding space and clustering analysis by molecular functions.



(a) Learned protein embedding space clustered by secondary-level biological processes.



(b) Silhouette scores by biological processes.

Figure 2.7: Visualization of learned protein embedding space and clustering analysis by biological processes.

2.4.2 Node type ablation study

This section aims to add an element of explainability to the protein representations from ProtGNN. To do this, I employed a node type ablation study to evaluate the node type importance, where I looked at which of the 9 node types in PrimeKG contributed the most to the protein representations. This involved training 9 separate GNN models, each excluding one of the 9 non-protein node types from the original graph. The same training hyperparameters, dataset split and model architecture outlined in Section 2.3 are used for each of the models. All models are finetuned on the MF-protein and BP-protein link-prediction task, except the MF-ablated model which is finetuned on only BP-link prediction and the BP-ablated model which is finetuned on only MF-link prediction. The sole difference between the 9 models is the removal of a single node type and all edges to these nodes from the knowledge graph. This allowed us to isolate and evaluate the contribution of each node type by comparing the resulting protein representations against those generated from the complete graph.

To evaluate the protein representations, I applied them to a downstream task of predicting associated enzyme commission (EC) numbers for each protein, which are critical for understanding enzymatic functions. More details about the implementation of downstream evaluation tasks are found in the next chapter (Section 3.3.4). By examining the performance differences in EC term prediction, the relative contribution of the omitted node type on the protein representations could be inferred. Results show that **molecular function** node type is the most important contributor to the protein representations for capturing functional properties (Table 2.1). While not as influential as the molecular function nodes, the next most important node types are pathway, effect/phenotype and drug. In addition, every ablated model performed worse on the EC prediction task than ProtGNN trained on the entire graph, indicating that all node types are needed and beneficial for function prediction.

Table 2.1: Downstream EC prediction F_{max} and AUPRC scores for protein representations from node type ablated models. Differences in metrics between each ablated model and ProtGNN are calculated. Sorted by Diff F_{max} .

Omitted Node Type	F_{max}	Diff F_{max}	AUPRC	Diff AUPRC
Molecular Function	0.7228	-0.0620	0.7180	-0.0687 ¹
Pathway	0.7610	-0.0238	0.7662	-0.0205 ³
Effect/Phenotype	0.7624	-0.0224	0.7740	-0.0127 ⁵
Drug	0.7641	-0.0207	0.7578	-0.0289 ²
Biological Process	0.7657	-0.0191	0.7740	-0.0127 ⁵
Disease	0.7704	-0.0144	0.7750	-0.0117 ⁷
Cellular Component	0.7701	-0.0147	0.7739	-0.0128 ⁴
Anatomy	0.7675	-0.0173	0.7798	-0.0069 ⁹
Exposure	0.7745	-0.0103	0.7791	-0.0076 ⁸
ProtGNN	0.7848	—	0.7867	—

2.4.3 Integrating protein sequence information with ESM embeddings

PrimeKG lacks any encodings of structural, molecular or sequential information regarding entity nodes such as proteins and drugs. Therefore, this section aims to evaluate whether sequence information for proteins can be injected into the model by using intentional node initializations rather than random Xavier initializations. I retrieved embeddings from Meta AI’s ESM2 model [5] which capture sequence information about the proteins and used them as protein node initializations. The aim is to see if this will improve the model’s performance on the link prediction task and whether the subsequent learned protein representations retain any of the original ESM embeddings from before training.

The same training method described in section 2.3.2 is followed, except before pretraining one model’s nodes are randomly initialized and a second model’s protein nodes are initialized as the corresponding ESM embeddings.

Embedding space evaluation

The following analyses will visualize the embedding spaces of the models, clustered and colored in different ways, to gain a deeper understanding of how the ESM initialized embeddings are effecting the GNN. I was first curious whether the final representations retained protein sequence information after training the model with ESM embeddings. To do this, I clustered the original ESM embeddings using k-means clustering and then visualized the embedding space of the original ESM embeddings, the final representations from the ESM model and the final representations from the random model. From the scatterplot of original ESM embeddings (Figure 2.8), the clusters from k-means can be clearly seen which serves as a verification of the clustering technique.

Looking at the embedding space for the ESM- and randomly-models colored by ESM clusters in Figure 2.9, most of the clustering information seems to be "forgotten" during the training process, as evident by the mostly random assortment of colors present. We note that some clusters do remain. For example cluster 6 (light sage green) and cluster 7 (red) are still together in the final representations. Looking at the gene names in green cluster 6, all proteins pertain to the family zinc finger proteins, and all proteins in red cluster 7 are olfactory receptors. However, because both of these clusters also appear in the randomly-initialized model (with no sequence information from ESM) we can conclude that information from PrimeKG identifies these protein families and not additional information from ESM. The other clusters, that dissipated after training, are made up of a diverse set of protein families upon preliminary analysis.

Given that the final embedding space for the ESM model didn't retain much of the protein sequence information, I was curious whether the ESM and random representations converged to a similar space. To test this theory, I clustered the final ESM model representations and subsequently visualized the final representations of the ESM and random models colored by those clusters. If the two models converged to similar embedding spaces, we'd expect to see that clusters found in one embedding space would also be found in the other. This is

the case, as seen in Figure 2.10. Both plots are colored based on a clustering of the final representations generated by the ESM initialized network. They share many of the same structures, indicating that the two models were able to pick up on the same features in the underlying information from the knowledge graph. Disregarding coloring and only looking at the shape and distance of clusters in the ESM-initialized representations, the ESM-initialized model better separates clusters. Therefore, the work that follows will use an ESM-initialized model, unless stated otherwise.

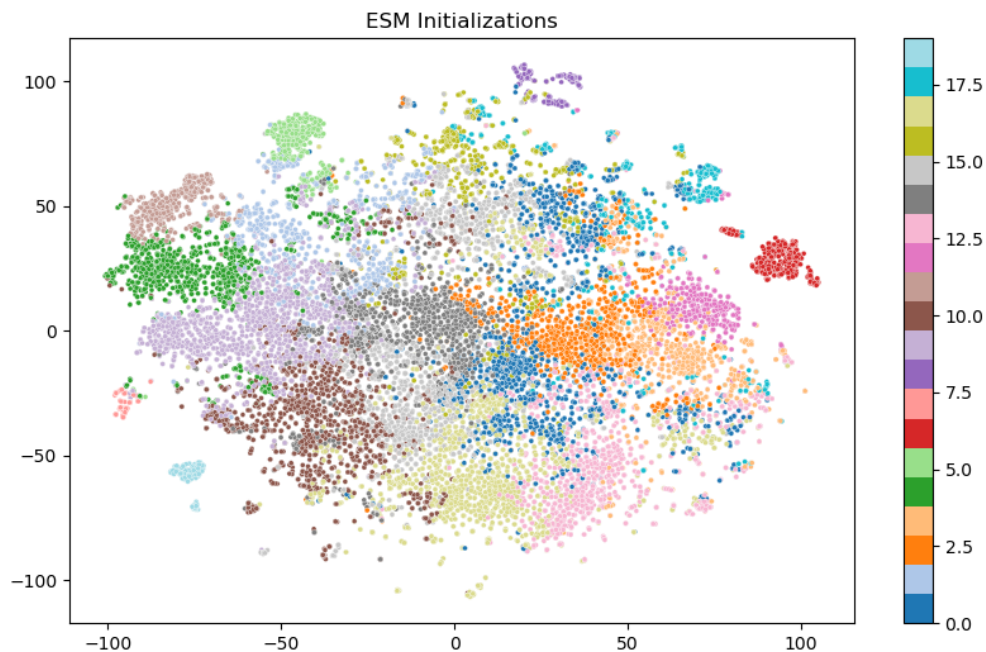


Figure 2.8: Embedding space for initial ESM representations clustered by k-means clustering.

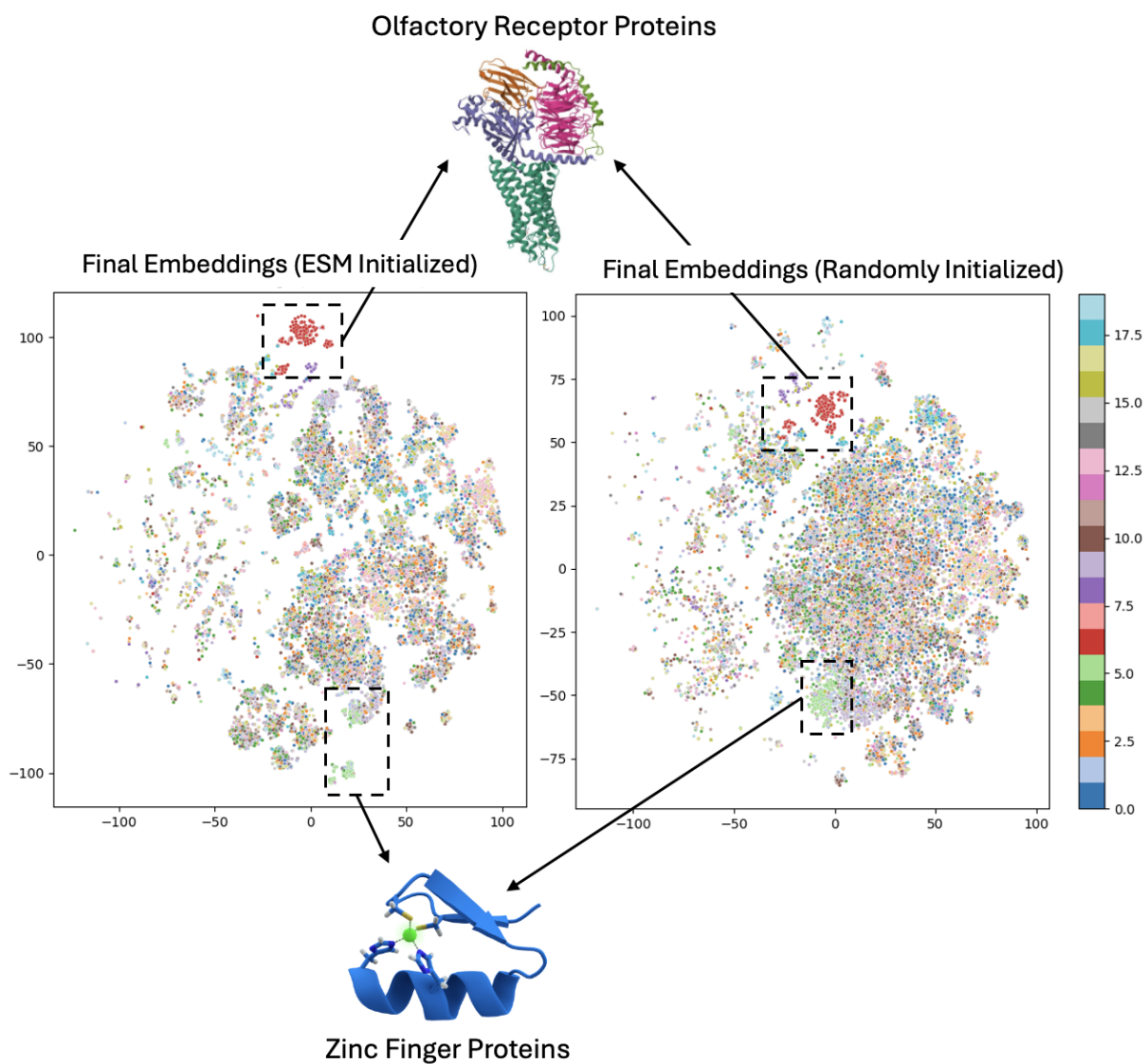


Figure 2.9: Protein embedding space for ESM and randomly initialized models colored by clusters from initial ESM representations.

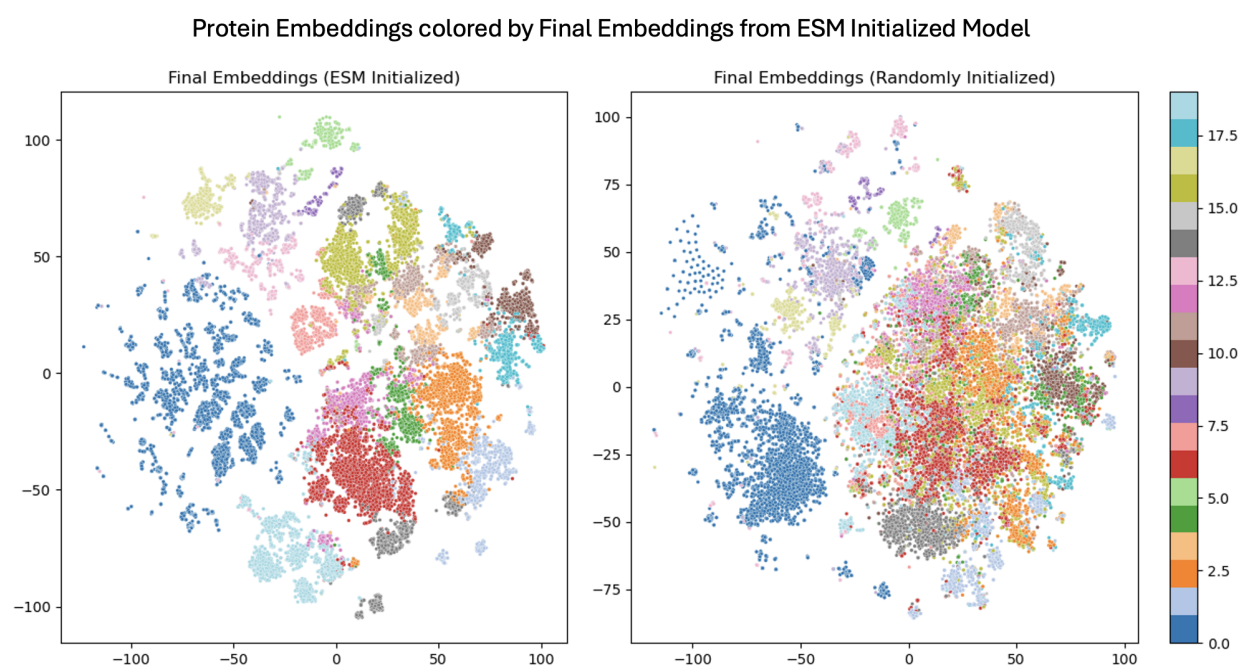


Figure 2.10: Protein embedding space for ESM and randomly initialized models colored by clusters found in the ESM-initialized model final embedding space.

Chapter 3

Structure to Function Model

3.1 Motivation

Protein function prediction remains a pivotal challenge in proteomics, with numerous models seeking to find functional insights from protein structures, because it’s known that protein structure is a determinant of protein function. GearNet is an example of such a model; directly predicting GO annotations based on structure alone [13]. This approach, while valuable, could benefit from a broader perspective by adopting a multi-hierarchical GNN model. I present GearNet-ProtGNN, which not only considers the structural elements of proteins but also integrates essential biomedical contextual information –providing a more comprehensive and accurate understanding of protein functions.

3.2 Prior Work

3.2.1 Baseline graph neural network

The baseline model used is GearNet-Edge (Geometry-Aware Relational Graph Neural Network) [13]. GearNet-Edge was developed to learn effective protein representations from structure graphs. The first part of the model is an encoder which learns geometric features

of proteins. The encoder is pretrained via multiview contrastive learning and self-prediction tasks, and then uses consequent representations for downstream protein property prediction tasks. The results showed that GearNet-Edge outperformed most state-of-the-art sequence based methods for protein representation and GO function prediction.

3.3 Methodology

My model, based on GearNet-Edge, uses the same structural representation for proteins and a relational, sparse-edge message passing algorithm which improves computationally efficiency.

3.3.1 Protein structure graph

Each protein is represented by a *residue-level* graph where each node is an alpha carbon. We can say that each protein has a graph, $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$. To construct a protein’s graph, there are 3 ways to add an edge:

1. **Sequential edges** – These edges connect two nodes within a sequential distance of 2. There are 5 edge types based on the distance between two nodes: $d \in \{-2, -1, 0, 1, 2\}$
2. **Radius edges** – Edges between nodes i and j are added if the Euclidean distance between the two is less than a set threshold, d_{radius}
3. **K-nearest neighbor edges** – A node is connected to it’s k-nearest neighbors based on Euclidean distance. This is to ensure similar edge density across proteins.

3.3.2 Model Architecture

GearNet-ProtGNN uses relational graph convolutional layers which are memory efficient but also have a large capacity for protein structure modelling [29]. For layer l and node i , the

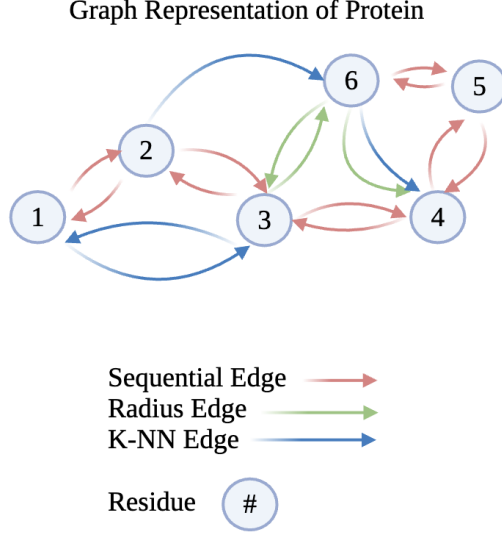


Figure 3.1: Example graph representation of protein structure.

corresponding representation is defined as $h_i^{(l)}$. Therefore, for each layer of message-passing, messages are aggregated from the neighboring nodes. $\mathcal{N}_r(i)$ is the set of neighboring nodes for node i with edge type r and $\mathbf{W}_r$ is the convolutional kernel for edge type r . BN denotes a batch normalization layer and $\sigma(\cdot)$ is the activation function, which is ReLU for my model.

$$\mathbf{u}_i^{(l)} = \sigma \left(\text{BN} \left(\sum_{r \in \mathcal{R}} \mathbf{W}_r \sum_{j \in \mathcal{N}_r(i)} \mathbf{h}_j^{(l-1)} \right) \right) \quad (3.1)$$

Lastly, each node’s representation is updated using the aggregation of messages from neighboring nodes and it’s own representation from the previous layer.

$$\mathbf{h}_i^{(l)} = \mathbf{h}_i^{(l-1)} + \mathbf{u}_i^{(l)} \quad (3.2)$$

In addition to the standard GCN layers, the model uses an edge-message passing layer to model the dependency between different edges of spatially close nodes. This method takes inspiration from AlphaFold2’s triangle attention in transformers [4]. A relational graph $\mathcal{G}' = (\mathcal{V}', \mathcal{E}', \mathcal{R}')$ is constructed where the nodes in $\mathcal{G}'$ are the edges from $\mathcal{G}$ and the edge type is the angle between the original edges. Then a similar R-GCN is applied to $\mathcal{G}'$ as described

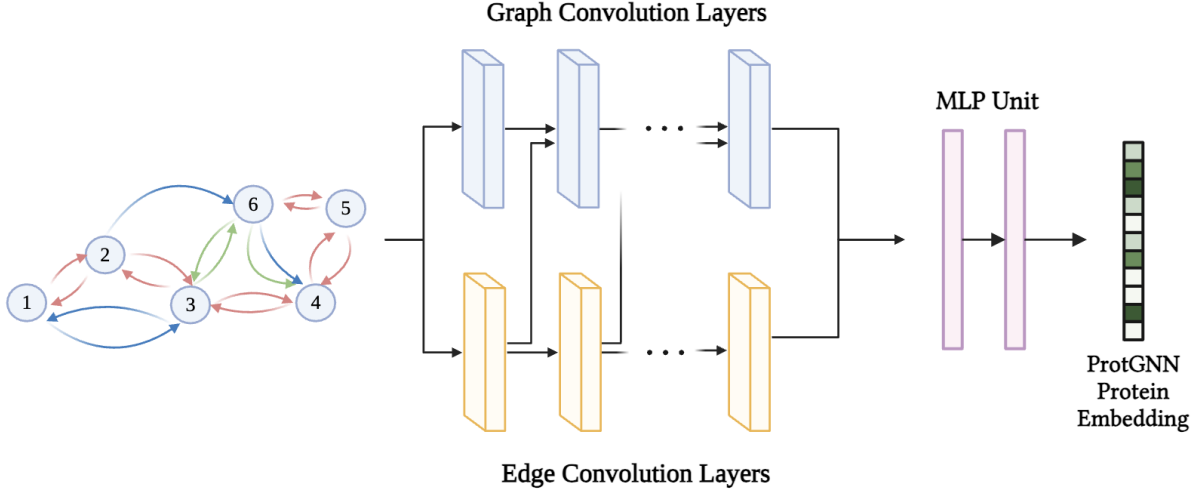


Figure 3.2: Architecture overview of relational GNN for predicting ProtGNN protein representations from structure graphs.

above. The update function for the R-GCN (Equation 3.1) is modified to Equation 3.3 such that the messages from the edge functions are included. Finally, the output of the GCN layers is fed into a 3-layer MLP projection head to make the final prediction.

$$u_i^{(l)} = \sigma \left(\text{BN} \left(\sum_{r \in \mathcal{R}} \mathbf{w}_r \sum_{j \in \mathcal{N}_r(i)} \left(\mathbf{h}_j^{(l-1)} + \text{FC}(\mathbf{e}_{i,j}^{(l-1)}) \right) \right) \right) \quad (3.3)$$

3.3.3 Training task and dataset

The objective for GearNet-ProtGNN is formulated as a supervised-learning regression task where the model predicts the 1024-dimensional learned protein representation from ProtGNN. For criterion, we use huber loss. Formally the loss function is

$$\mathcal{L}_i = \text{HUB}(f(g_i), p_i) = \begin{cases} \frac{1}{2}(f(g_i) - p_i)^2 & \text{if } |f(g_i) - p_i| \leq \delta, \\ \delta (|f(g_i) - p_i| - \frac{1}{2}\delta) & \text{otherwise.} \end{cases} \quad (3.4)$$

where $f(g_i)$ is the predicted protein representation and p_i is the target protein representation from ProtGNN.

I used the GeneOntology dataset from TorchDrug [30] for training which contains PDB files and GO annotations for 34,592 proteins. A new dataset loader was implemented to retrieve the corresponding protein representations from ProtGNN for the proteins in the TorchDrug GO dataset. Proteins from PrimeKG are mapped to proteins in the TorchDrug GO dataset by converting the PDB ID to the corresponding UniProt accession name and then to the gene name. Note that the mapping from PDB ID to gene name is not one-to-one; multiple proteins in the TorchDrug GO dataset are subdomains of the same gene product or are structures from other species but for the same protein. Therefore multiple proteins in the TorchDrug GO dataset may share the same target representations from ProtGNN. The final dataset size after mapping is 10,341 unique protein structures and 4,813 unique target protein representations. For the dataset split, I implement the multi-cutoff split method in Gligoričević et al. [2] to make sure that there is not more than 95% sequence identity between proteins in the training and the test sets.

Model hyperparameters were optimized by a hyperparameter sweep conducted in Weights & Biases to optimize prediction of protein target embeddings. Optimized hyperparameters can be found in Appendix B.2.

3.3.4 Downstream protein function prediction tasks

I adopted four downstream tasks proposed in the GearNet paper [13]. The first three are GO term prediction, which aim to predict which GO terms a protein is annotated with. This downstream task is formulated as a multi-class binary prediction problem where each GO term is its own class. GO terms from all three ontologies are predicted (molecular function, biological processes and cellular components). The fourth downstream task is Enzyme Commission (EC) number prediction, which is a classification scheme based on the biochemical reactions they catalyze.

For these downstream tasks, an additional 3-layer MLP projection head is appended after the R-GCN layers. I tested whether freezing the R-GCN layers improved model per-

Table 3.1: Number of binary prediction tasks for each annotation type.

Annotation Type	Number of Tasks
GO-MF	489
GO-BP	1943
GO-CC	320
EC	538

formance. For some tasks, performance improved and for others it did not. The best model between frozen and unfrozen was used for comparison (specified in Table 3.3). For ProtGNN evaluation, the protein representations are directly fed into a 3-layer MLP network instead. The evaluation metric used for all downstream prediction tasks is F_{max} and AUPRC.

3.3.5 Testing model performance for different protein truncation lengths

The choice of maximum truncation length for truncating amino acid sequences is a critical factor in model design, influencing its ability to capture the structural components of proteins. A lower cutoff, such as 100 amino acids, may be beneficial as it focuses on the most functionally significant regions of proteins, like active sites or key motifs. This could lead to a more efficient representation of the critical aspects of protein function while reducing computational complexity and noise. On the other hand, not truncating sequences allows for a comprehensive analysis of the entire protein, including all functional domains affecting its overall conformation and behavior. For reference, the average length of a protein is 472/430 amino acids [31]. By testing 4 truncation schemes (100, 350, 600, None), I identified the optimal threshold that balanced computational efficiency and structural representation, and thereby optimizing the predictive power of GearNet-ProtGNN. The four models are evaluated based on their validation huber loss and F_{max} for downstream GO-MF prediction. Based on the results in Table 3.2, the optimal truncation threshold is 600 amino acids. Assume for all models described here on out use this threshold value, unless stated otherwise.

Table 3.2: Model evaluation results for different truncation thresholds.

Truncation Threshold	Valid Huber Loss	F_{max} GO-MF
100 AA	0.2026	0.4403
350 AA	0.2001	0.5147
600 AA	0.2001	0.5299
None	0.2023	0.5189

3.3.6 Retrieving protein structural families

For each protein in the TorchDrug GO dataset, I retrieved the CATH superfamily. The CATH database [32], developed at the University College London, captures hierarchical relationships between protein domains. Protein domains from PDB [33] are classified within the CATH hierarchy: Class (C), Architecture (A), Topology (T) and Homologous Superfamily (H). Class captures a protein’s secondary structure content (mainly alpha, mainly beta, or a mixture). Architecture captures information about the 3D arrangement of secondary structures in space (sandwich, roll, etc.). Topology captures how the secondary structures are connected. Finally, homologous superfamily groups together domains that are related by evolution and function. For all analyses looking at how the protein embedding space captures structural information, only the CAT classification for each protein is used, because the H domain includes functional information and I was interested in purely structure.

3.4 Evaluation

3.4.1 Downstream prediction task results

For the downstream prediction tasks, I compared my model, GearNet-ProtGNN, to the baseline model GearNet-Edge. The main difference is that GearNet only encodes structural information, whereas GearNet-ProtGNN uses a multi-hierarchical GNN training scheme to encode both biomedical context and structural information. In addition, I evaluate the

ProtGNN model discussed in Chapter 2, which encodes biomedical information only.

The results of the downstream prediction tasks are reported in Tables 3.3 and 3.4. Our multi-modal model outperforms the structure-only model on all 4 of the datasets for both F_{max} and AUPRC metrics. By comparing the bottom three rows, we find that GearNet-ProtGNN obtains much better results for EC, GO-MF and GO-CC datasets and slightly better results for GO-BP. Incorporating ESM sequence information did not significantly improve GearNet-ProtGNN. Overall, this shows the power of using a multi-modal model and that structure is not the only factor in function prediction.

Notably, ProtGNN achieves the highest F_{max} scores across multiple categories (EC, GO-BP, and GO-CC prediction tasks) as indicated by the asterisks. However, it’s important to note that the model was trained on a graph containing molecular function, biological process and cellular component nodes. Since these nodes are directly related to the target annotations of the downstream tasks, there could be potential for data leakage and therefore inflated performance metrics.

Table 3.3: F_{max} from EC and GO downstream prediction tasks. The bold numbers indicate the best result for each task. [*] While results are shown for ProtGNN representations and they had the highest score for these downstream tasks, there is potential data leakage, and therefore not considered for the best result. [†] denotes R-GCN layers were frozen.

Model	EC	GO-BP	GO-MF	GO-CC	Information Encoded
ProtGNN	0.7848*	0.4104*	0.4260	0.5381*	Biomedical Context
ProtGNN-ESM	0.7175	0.3966	0.3948	0.5175	Biomedical Context Sequence
GearNet-ProtGNN	0.7563	0.3647 [†]	0.5257	0.3916 [†]	Biomedical Context Structure
GearNet-ProtGNN-ESM	0.7676	0.3378	0.5197	0.4047	Biomedical Context Structure Sequence
GearNet-Edge	0.6892 [†]	0.3423	0.4584 [†]	0.3548	Structure

Table 3.4: **AUPRC** from EC and GO downstream prediction tasks. The bold numbers indicate the best result for each task. [*] While results are shown for ProtGNN representations and they had the highest score for these downstream tasks, there is potential data leakage, and therefore not considered for the best result. [†] denotes R-GCN layers were frozen.

Model	EC	GO-BP	GO-MF	GO-CC	Information Encoded
ProtGNN	0.7867*	0.3215*	0.4344	0.4257*	Biomedical Context
ProtGNN-ESM	0.7112	0.3161	0.4106	0.3897	Biomedical Context Sequence
GearNet-ProtGNN	0.7700	0.2498 [†]	0.5288	0.2765 [†]	Biomedical Context Structure
GearNet-ProtGNN-ESM	0.7830	0.2349	0.5251	0.2423	Biomedical Context Structure Sequence
GearNet-Edge	0.7132 [†]	0.2323	0.4790 [†]	0.2348	Structure

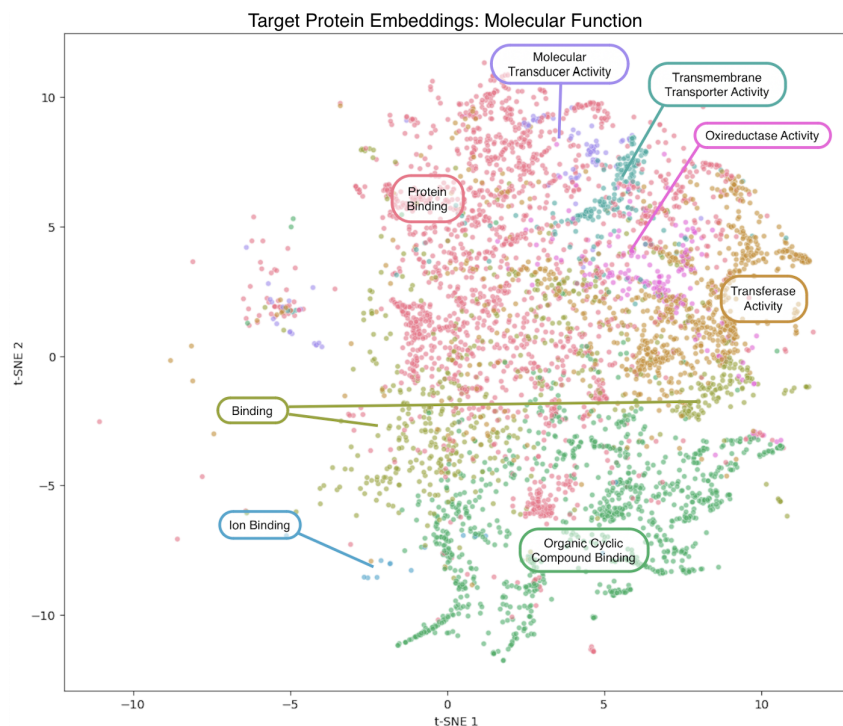
3.4.2 Visualizing the predicted protein embedding space

In conducting a qualitative analysis of GearNet-ProtGNN, my focus was on comparing the predicted embedding space with the target embedding space from ProtGNN. Specifically, I investigated the clustering efficacy of the predicted embedding space in relation to enriched GO annotations: does it cluster as effectively, better, or worse than the target representations? The methods used for clustering and labelling were consistent with those in Section 2.4.1. For dimensionality reduction, because t-SNE is an iterative algorithm that is fit for a given set of vectors, it is not possible to compute a t-SNE reduction transformation on the target representations and then apply it to the predicted representations. Therefore, I computed the dimensionality reduction transformation on both the target and predicted representations within the same t-SNE process, ensuring that they maintain their relative proximities and ability to be compared across separate plots. Additionally, only the proteins in the Torch GO dataset (see Section 3.3.3) are looked at and their corresponding representations from ProtGNN. This is why the target embedding space in Figure 3.4a and 3.3a appears sparse on the left side, where the other representations from ProtGNN not used to train GearNet-ProtGNN would map to.

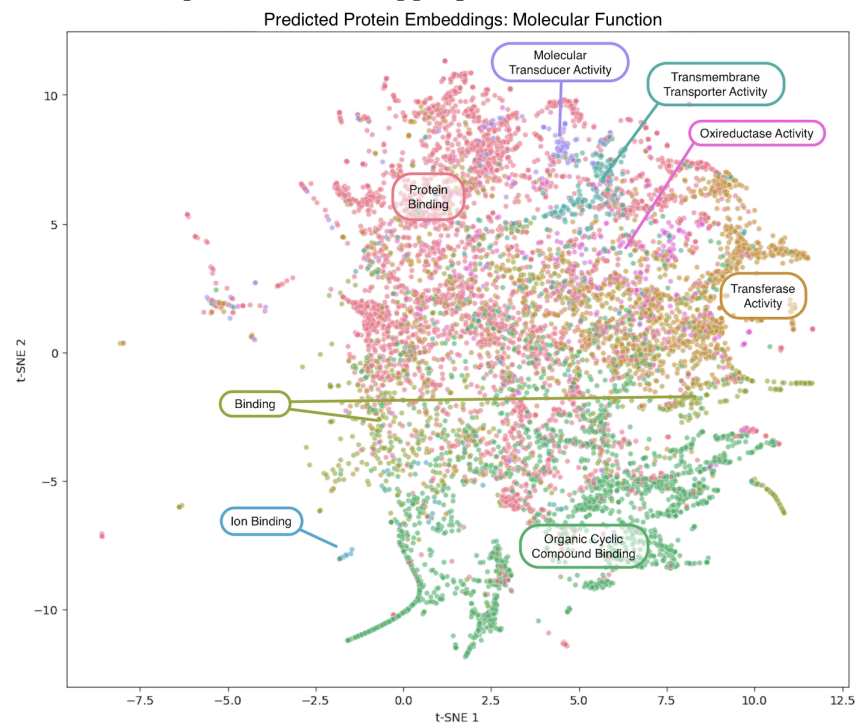
The similarity in coloring between the predicted protein embedding space and the target protein embedding space show that GearNet-ProtGNN is able to accurately replicate the target embedding space that encodes biomedical information. Looking first at Figure 3.3, and observing the shape and distribution while disregarding color, it is clear that a protein maps to a similar position between the predicted and target embedding spaces, showcasing GearNet-ProtGNN’s ability to faithfully reconstruct the target embedding space. Now observing the colors representing enriched molecular function groups, the predicted embedding space can be clustered by function accurately, in almost the exact same position in the space. The same analysis is done but colored by enriched biological processes in Figure 3.4. In addition, I calculated the mean silhouette coefficients for the predicted, target and baseline GearNet-Edge embedding spaces for each label type: molecular function and biological process (Table 3.5). As expected, the original target embedding space – from ProtGNN trained on PrimeKG containing MF and BP nodes – clusters the best by these labels, followed closely by the predicted embedding space. GearNet-Edge clusters significantly worse by GO annotations, which makes sense as it is trained only on structure.

Table 3.5: Mean silhouette coefficients for different embedding spaces and label types.

Embedding Space	Label Type		
	BP	MF	Structure
GearNet-ProtGNN (Predicted)	-0.0776	-0.0608	-0.0459
ProtGNN (Target)	-0.0310	-0.0041	-0.0225
GearNet-Edge	-0.2713	-0.2749	-0.1749

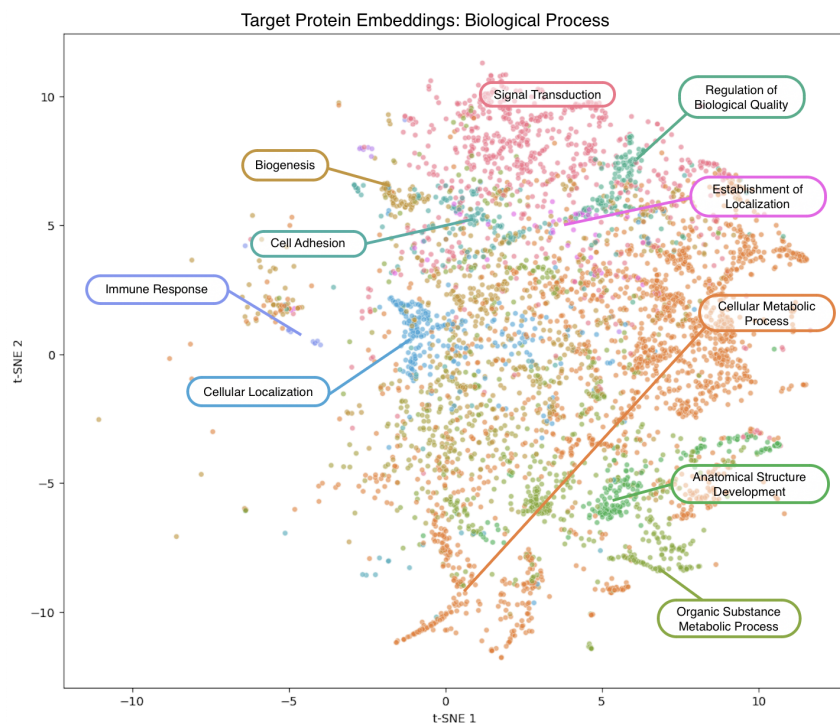


(a) Target protein embedding space from ProtGNN colored by secondary-level molecular functions. Only viewing proteins with representations mapping to a PDB structure in TorchDrug GO dataset.

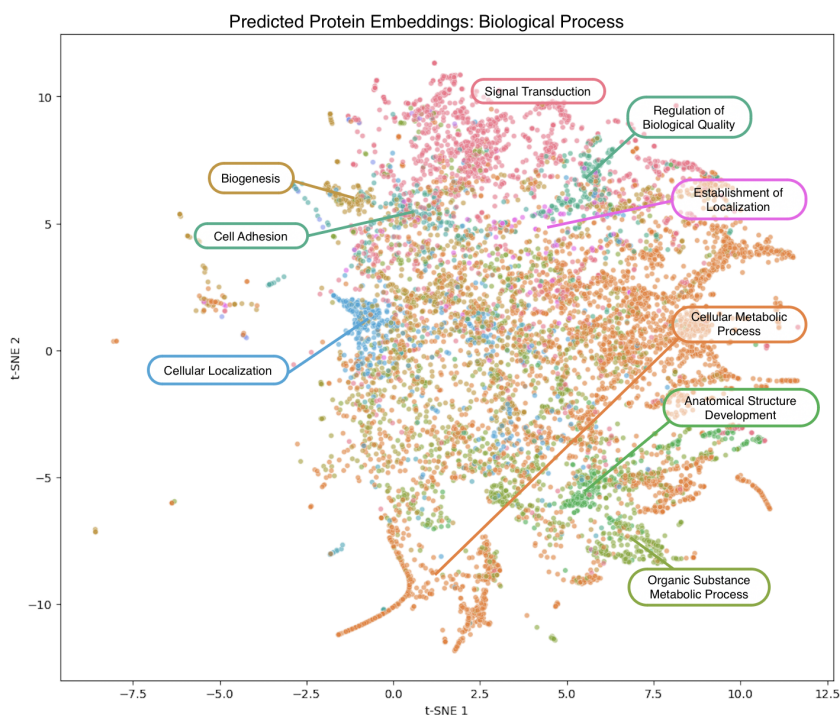


(b) Predicted protein embedding space from GearNet-ProtGNN, colored by secondary-level molecular functions.

Figure 3.3: Visualization of predicted embedding space compared to target embedding space clustered by molecular functions.



(a) Target protein embedding space from ProtGNN clustered by secondary-level biological processes. Only viewing proteins with representations mapping to a PDB structure in the TorchDrug GO dataset.



(b) Predicted protein embedding space from GearNet-ProtGNN colored by secondary-level biological processes.

Figure 3.4: Visualization of predicted embedding space compared to target embedding space clustered by biological processes.

To evaluate how well the predicted embeddings are for PDB structures in the TorchDrug GO dataset that are domains of the same protein and therefore share target ProtGNN representations (see Section 3.3.3), I plotted the predicted embedding space and highlighted 10 groups of proteins, where within each group all proteins have the same target representation. Figure 3.5 shows that for the most part GearNet-ProtGNN is able to repredict similar representations for proteins that share the same target representation. This further proves the convergence of the GearNet-ProtGNN protein embedding space to the target ProtGNN embedding space.

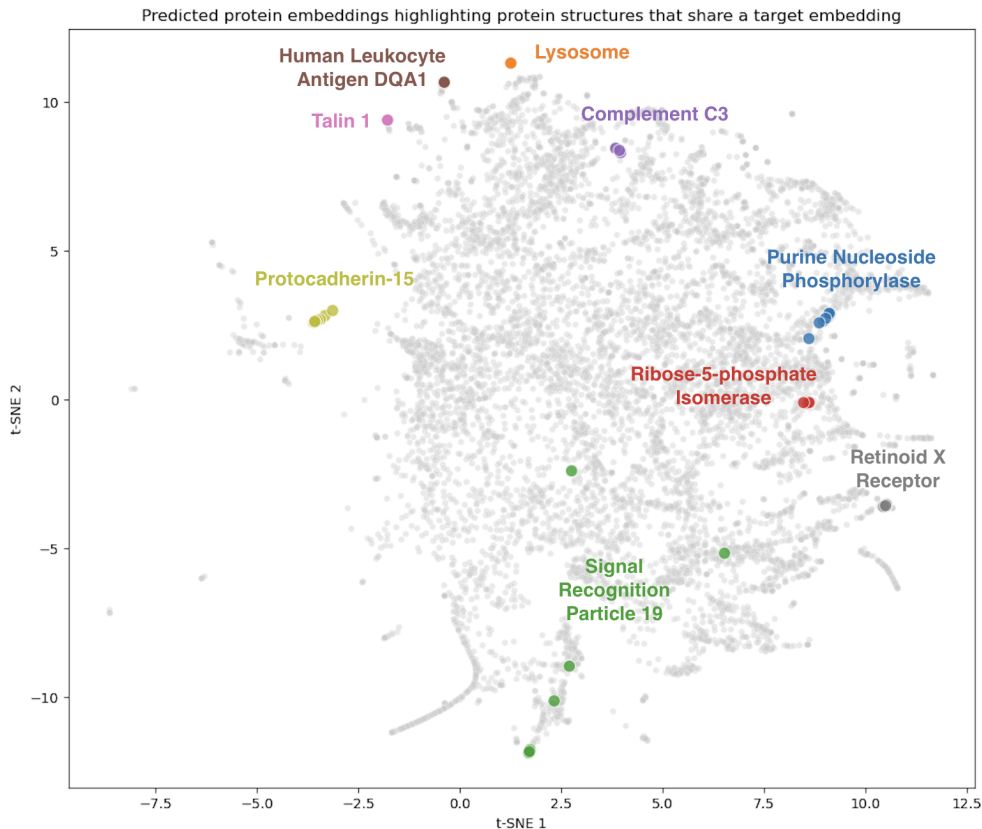


Figure 3.5: GearNet-ProtGNN predicted embedding space highlighting groups of proteins that share the same ProtGNN target representations.

Furthermore, to determine whether the predicted embedding space encodes structural information, I explored clustering by enriched protein structure families. The CAT(-H) classifications for all proteins were retrieved as described in Section 3.3.6 and then I colored the embedding space by the top-10 CAT superfamilies. All other proteins were classified as

"Other". Looking at Figure 3.7, we see that certain structural families cluster well in the protein embedding space – Alpha-Beta 2-layer sandwich phosphorylase kinase (light green) and mainly Alpha orthogonal bundle arc repressor mutant (pink). Figure 3.6 supports this, where these 2 structural families have positive mean silhouette coefficients (0.119 and 0.052, respectively). This structural encoding analysis affirms the fact that there is a strong link between a protein’s structure and function. Interestingly, Alpha-Beta 3-layer sandwich glutaredoxin (dark gray) has a relatively high silhouette coefficient (0.058) but is not clearly clustered in Figure 3.7.

Finally, it is surprising that the target embedding space clustered better by structural families than the predicted embedding space (GearNet-ProtGNN) and the GearNet-Edge latent space both generated by models trained on protein structure graphs (see Table 3.5). Future analysis needs to be done to determine why this is the case.

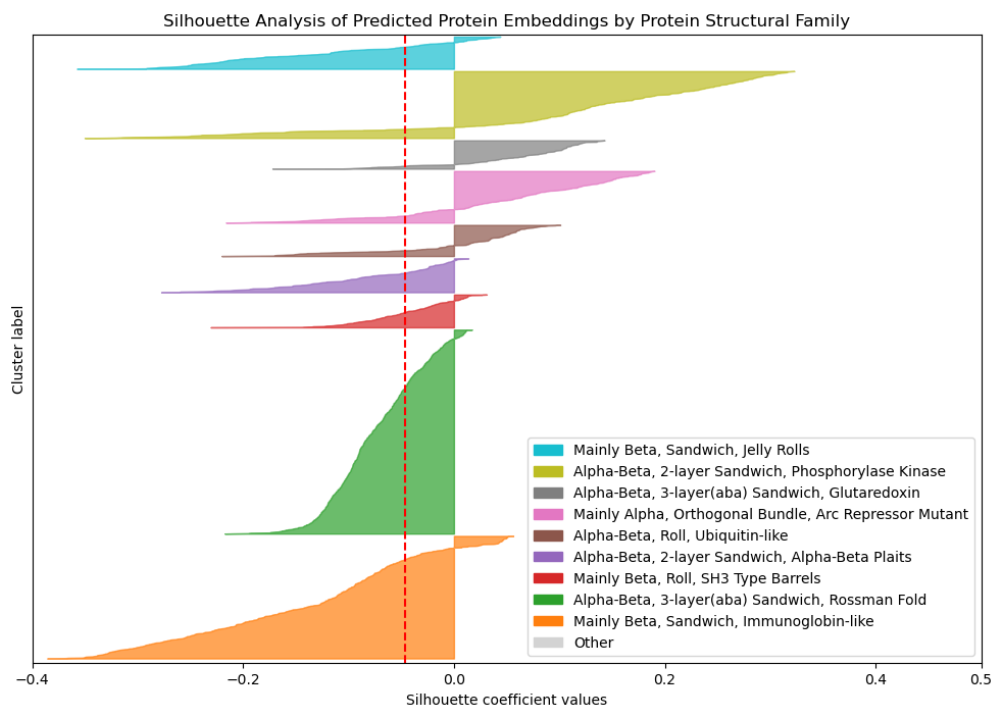


Figure 3.6: Silhouette scores for the GearNet-ProtGNN predicted embedding space by enriched structural families.

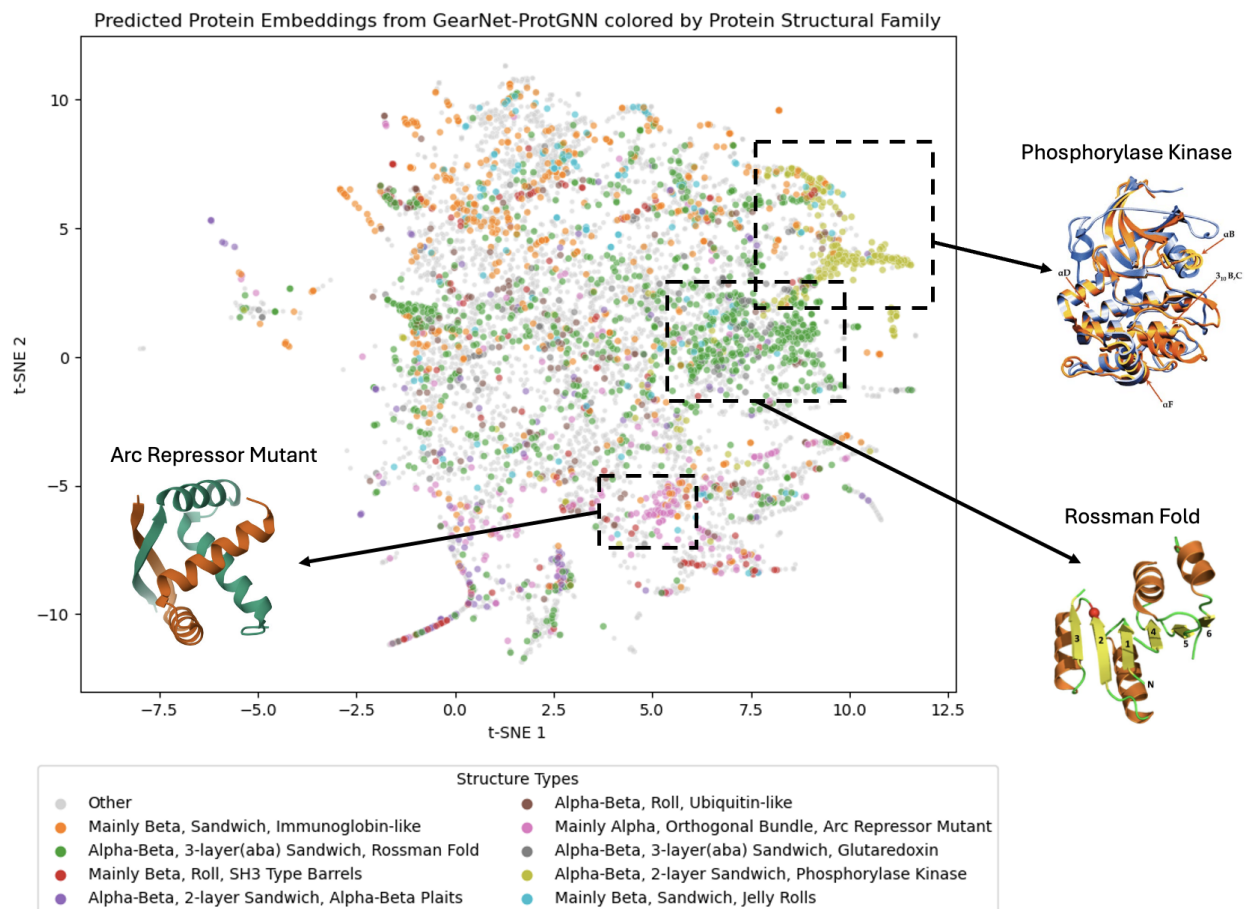


Figure 3.7: GearNet-ProtGNN predicted embedding space colored by enriched structural families for the top 10 CAT superfamilies.

Chapter 4

Discussion

4.1 Key findings

In the discussion section, I'll begin by highlighting the capabilities of ProtGNN in learning representations that encode the biomedical context of proteins using a relational GNN. The key findings regarding ProtGNN include:

1. **Learning Biomedical Encodings:** ProtGNN is able to learn representations that encode detailed biomedical information relevant to proteins, due to its use of a relational GNN architecture integrating diverse biological data types and understands their relationships through the message-passing layers.
2. **Clustering by Functional Annotations:** The learned protein representations cluster well based on molecular function and biological process annotations. Specifically, smaller functional groups display tighter clustering, indicating a strong encoding of the protein's specific functions in its representation.
3. **Influence of Molecular Function Node Types:** The molecular function node type in PrimeKG significantly contributes to the final learned protein representations that optimized BP-protein and MF-protein link prediction.

4. **No Retention of Sequence Information:** The integration of ESM sequence information into ProtGNN during node initialization improves the separability of clusters found in the embedding space. However, the final representations lost all sequence-related information. Future work needs to be done to better integrate the ESM embeddings, potentially through concatenating them after every convolutional layer or using feed-forward connections.

Building on the foundation set by ProtGNN, I introduced GearNet-ProtGNN, the second module of a novel multi-modal and multi-hierarchical GNN model that integrates three modalities of information regarding proteins to achieve state-of-the-art performance on protein function prediction. Unlike traditional approaches that might use an alignment module to train multiple deep learning models, GearNet-ProtGNN trains two GNNs sequentially targeting the embedding space from the first GNN in the second GNN. This method is not only computationally more efficient but is also accurate, showing its robustness. Key achievements of GearNet-ProtGNN include:

1. **High Performance in Protein Property Prediction:** When used for downstream prediction tasks such as predicting EC numbers and GO annotations, GearNet-ProtGNN consistently outperforms models that only use structural encoders. This highlights the effectiveness of integrating multiple modalities of protein information.
2. **Reconstruction of ProtGNN Embedding Space:** Through visual analysis, GearNet-ProtGNN can faithfully reconstruct representations that encode a range of biomedical information based on the protein structure graph.
3. **Encoding of Protein Structure Information:** GearNet-ProtGNN representations can also effectively group proteins based on their structural families. This clustering shows that the representations generated by the model accurately capture and reflect structural properties, which are important for the prediction of protein function.

4.2 Limitations

One of the limitations of the downstream task evaluation technique implemented here is there is potential for training data leakage. The model – GearNet-ProtGNN – was trained to predict ProtGNN representations from protein structure. However, the ProtGNN representations inherently encode some information about GO terms because GO terms are nodes in PrimeKG from which ProtGNN was trained on. There are two possible directions to take to prevent this. One, ProtGNN could be retrained on the biomedical knowledge graph with molecular function, biological process and cellular component nodes omitted. Two, the model could be evaluated on other downstream tasks. EC number prediction is still a good downstream task as EC numbers are not in PrimeKG, and others could include protein-protein interaction prediction.

Another potential limitation is how the ProtGNN representations were evaluated during the downstream tasks. Because the representations were directly fed into a 3-layer MLP network, compared to the other models evaluated whose weights from their respective R-GCN layers were loaded and prepended to a 3-layer MLP head, the MLP has a larger role during training. Therefore this may not be a fair comparison between ProtGNN and GearNet-ProtGNN. A solution would be to rework the ProtGNN codebase and include the relational GNN layers from ProtGNN before a MLP head.

Finally, because the training objective for GearNet-ProtGNN is to reconstruct the ProtGNN embedding space, there is more emphasis on retaining the biomedical domain information than on retaining structural information. While analyses showed that the representations from GearNet-ProtGNN do encode some structural information, it may not be optimal. Future work could be done to focus on improving the model architecture, such as adding skip connections or having two training objectives including structure graph reconstruction, to ensure a good balance between biomedical context and structure information in the learned protein representations.

Chapter 5

Future Work

I’ve introduced a successful framework to integrating multi-modal data into a single protein embedding space that can effectively be used to predict protein function. Besides future directions to improve the model and address its limitations which were discussed in Chapter 4, there remains many possible future applications to extend the use cases of such a diverse, rich and informative embedding space.

5.1 Interactive visualization of the protein space

The current visual analysis is a static 2D representation of the embedding space which implements a definite approach to assigning protein function labels, in order to see if the embedding space captures functional encodings on a high level. However, in reality, proteins have multiple functions at varying hierarchical levels. It would be interesting to have a dynamic way of assigning protein labels, allowing us to view the embedding space at different levels of granularity. Additionally, rather than using this embedding space as a protein representation for property prediction tasks, it could be leveraged as an interactive tool for biologists and researchers to explore the protein domain. One potential method of protein function labelling and ontology generation would be to employ advanced large language models. By parsing scientific literature, such as abstracts and UniProt/Swiss-Prot descriptions

associated with each protein, functional information can be extracted and synthesized to be used for clustering. This process would use contextual analysis and dynamic clustering to adapt ontology as new data becomes available, creating a responsive labelling system. In addition, graph-based analysis techniques could be used, including weighted analysis of protein neighbors in the embedding space to infer function through first-, second-, and third-degree associations. While this is outside the scope of the work for my thesis and thus the details are not described here, undergraduate students in the Kellis lab have begun working on such a visualization by using the embedding space from GearNet-ProtGNN and protein descriptions extracted from UniProt/Swiss-Prot (Mantis AI). An interactive, user-friendly visualization tool would allow researchers to explore these complex relationships dynamically, with capabilities for user feedback to refine and update the model and ontology.

5.2 Incorporation into protein large language models

The integration of the protein embedding space from ProtGNN into a protein-specific large language model (LLM) represents a promising downstream application, specifically through the use of these representations as outbeddings. In such a setup, the representations would be used to generate textual descriptions of proteins, effectively translating complex molecular features into comprehensive, human-readable text. This method leverages the detailed structural and biomedical-related data encoded in the representations to produce accurate and relevant descriptions of protein characteristics, interactions, and functions. Such a system could improve the accessibility of protein data, making it invaluable for researchers who require quick summaries of protein functions without delving into detailed structural data. GPT models are a good example of an outbedding-to-text framework. Models like OpenAI’s GPT series [8] are trained to predict the next word in a sequence, effectively transforming embeddings into textual outputs. A similar architecture could be applied to this smaller, yet-rich embedding space of proteins.

5.3 Application to drug discovery

A similar framework to the one I’ve outlined in my thesis could be applied to get a therapeutic-centric embedding space. Because drugs are already present in the biomedical knowledge graph used, PrimeKG, a GNN could be finetuned on drug-disease and all drug-X links (where X is any node type) as done in TxGNN [23]. Then these drug representations could be similarly used as targets for a second GNN that predicts where a drug falls in this latent space from structure. Rather than using amino acid structure as input, we’d use the drug molecular structure derived from SMILES.

There’s a vast need for a well-defined drug embedding space for downstream prediction tasks that could speed up the therapeutic discovery pipeline. Such a space could be applied to drug repurposing, where embedding spaces can identify similarities between drugs based on the molecular structure, mechanism of action, effects on biological pathways and diseases, all of which are captured in the embedding space. This can then highlight potential new uses for existing drugs, speeding up the process of finding treatments for diseases by bypassing some of the early stages of drug development. Another potential application is predicting drug interaction. By understanding the proximity of drugs in a embedding space, researchers could anticipate adverse reactions or synergistic effects, which is crucial for the safety and effectiveness of polypharmacy. Similar to predicting interactions, representations can help forecast possible side effects by analyzing the characteristics of drugs that share similar side effect profiles in the embedding space. Or by analyzing how drugs cluster around certain biological targets or pathways, it could be possible to identify and validate new targets for drug development. This can lead to the discovery of novel therapeutic strategies for diseases that are currently difficult to treat. These use cases illustrate how a well-constructed drug-centric embedding space from a multi-modal double-GNN framework could be a transformative tool in pharmaceutical research and development.

Appendix A

Code Availability & Compute Resources

The code for **ProtGNN** to reproduce results with documentation is publicly available at <https://github.com/emmatysinger/ProtGNN>.

PrimeKG can be downloaded from [Harvard Dataverse](#)¹.

The code for **GearNet-ProtGNN** to reproduce results with documentation is publicly available at <https://github.com/emmatysinger/GearNet-ProtGNN>.

Details about importing **TorchDrug GO dataset** can be found on [TorchDrug API Documentation](#)².

All models were trained on A100 Tensor Core GPUs on [Openmind](#), a shared computing cluster hosted by the McGovern Institute for Brain Research³.

¹PrimeKG: <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/IXA7BM>

²TorchDrug API Documentation URL: <https://torchdrug.ai/docs/api/datasets.html#geneontology>

³Openmind URL: <https://openmind.mit.edu/>

A.1 Dataset statistics

Table A.1: Dataset statistics for ProtGNN, GearNet-ProtGNN, and downstream tasks.

Dataset	Unit	Model	Split Size		
			Train	Test	Valid
PrimeKG	Edges	ProtGNN	6,733,536	405,024	961,938
TorchDrug GO	Proteins	GearNet-ProtGNN	7,639	1,817	885
Downstream EC Pred.	Proteins	–	2,968	390	379
Downstream GO Pred.	Proteins	–	7,077	1,585	786

Appendix B

Model Hyperparameters

B.1 ProtGNN

Table B.1: Final hyperparameters used for pretraining and finetuning ProtGNN. [*] denotes a hyperparameter that was optimized during the hyperparameter tuning sweep. [†] denotes a hyperparameter value taken from [23].

Hyperparameter		Pretraining	Finetuning
GNN	# of layers	6 [†]	6 [†]
	input dim.	512	512*
	hidden dim.	512	512*
	output dim.	1024	1024*
	dropout	0.1 [†]	0.1 [†]
Learning	optimizer	AdamW	AdamW
	learning rate	1e-3*	1e-3*
	weight decay	0 [†]	0 [†]
	batch size	512*	—
	# epoch	2*	50*

B.2 GearNet-ProtGNN

Table B.2: Final hyperparameters used for training GearNet-ProtGNN. [*] denotes a hyperparameter that was optimized during the hyperparameter tuning sweep. [[†]] denotes a hyperparameter value taken from [13].

	Hyperparameter	GearNet-ProtGNN
GNN	# of layers	6 [†]
	hidden dim.	512 [†]
	dropout	0 [†]
MLP	# of layers	5*
	hidden dim.	3072
	output dim.	1024
	dropout	0.3
	activation	ReLU
Learning	optimizer	Adam
	learning rate	1e-3*
	weight decay	0 [†]
	batch size	32*
	# epoch	100*

Table B.3: Hyperparameters for evaluating models on the downstream tasks: GO and EC prediction.

	Hyperparameter	GO	EC
MLP	# of layers	3	3
	hidden dim.	1024	1024
	dropout	0.3	0.3
Learning	optimizer	Adam	Adam
	learning rate	1e-5	1e-5
	weight decay	1e-4	1e-4
	batch size	24	24
	# epoch	25	25

Appendix C

Expanded Results

C.1 Link-prediction task evaluation

Table C.1: Validation AUROC, AUPRC and Huber loss for different finetuning link-prediction tasks.

Link Prediction Task	AUROC	AUPRC	Loss
Protein-MF	0.9038	0.9031	0.4084
Protein-BP	0.8827	0.8836	0.4669
Protein-MF & Protein-BP	0.8873	0.8870	0.4464

Table C.2: Downstream EC prediction F_{max} and AUPRC scores for protein representations from different finetuning link-prediction tasks.

Link Prediction Task	F_{max}	AUPRC
Protein-MF	0.4635	0.4411
Protein-BP	0.4675	0.4214
Protein-MF & Protein-BP	0.4794	0.4356

C.2 Silhouette scores evaluation

Table C.3: Mean silhouette coefficients by molecular function for ProtGNN.

Molecular Function	Mean Silhouette Coef.
Ion binding	0.109
Transferase activity	0.062
Catalytic activity, acting on a protein	0.036
Transmembrane transporter activity	0.013
Molecular transducer activity	0.007
Binding	-0.013
Oxidoreductase activity	-0.012
Organic cyclic compound binding	-0.076
Protein binding	-0.121
All	-0.049

Table C.4: Mean silhouette coefficients by biological process for ProtGNN.

Biological Process	Mean Silhouette Coef.
Detection of stimulus	0.367
Immune response	0.159
Cell death	0.144
Vesicle-mediated transport	0.110
Cellular developmental process	0.082
Anatomical structure development	0.039
Nitrogen compound metabolic process	0.035
Regulation of biological quality	0.034
Establishment of localization	-0.018
Cell adhesion	-0.039
Cellular metabolic process	-0.053
Signal transduction	-0.072
Biogenesis	-0.083
All	-0.029

Table C.5: Mean silhouette coefficients by protein structure for GearNet-ProtGNN.

Protein Structure	Mean Silhouette Coef.
Alpha-Beta, 2-layer Sandwich, Phosphorylase Kinase	0.119
Alpha-Beta, 3-layer Sandwich, Glutaredoxin	0.058
Mainly Alpha, Orthogonal Bundle, Arc Repressor Mutant	0.052
Alpha-Beta, Roll, Ubiquitin-like	0.004
Mainly Beta, Roll, SH3 Type Barrels	-0.046
Alpha-Beta, 3-layer Sandwich, Rossmann Fold	-0.074
Alpha-Beta, 2-layer Sandwich, Alpha-Beta Plaits	-0.085
Mainly Beta, Sandwich, Jelly Rolls	-0.123
Mainly Beta, Sandwich, Immunoglobulin-like	-0.137
All	-0.046

References

- [1] C. J. Jeffery, “Current successes and remaining challenges in protein function prediction,” *Front. Bioinform.*, vol. 3, Jul. 2023.
- [2] V. Gligorijević, P. D. Renfrew, T. Kosciolk, *et al.*, “Structure-based protein function prediction using graph convolutional networks,” *Nat. Commun.*, vol. 12, no. 1, 2021.
- [3] M. Varadi, S. Nair, I. Sillitoe, *et al.*, “3D-Beacons: decreasing the gap between protein sequences and structures through a federated network of protein structure data resources,” *GigaScience*, vol. 11, Nov. 2022. DOI: [10.1093/gigascience/giac118](https://doi.org/10.1093/gigascience/giac118).
- [4] J. Jumper, R. Evans, A. Pritzel, *et al.*, “Highly accurate protein structure prediction with AlphaFold,” *Nature*, vol. 596, no. 7873, pp. 583–589, Aug. 2021.
- [5] Z. Lin, H. Akin, R. Rao, *et al.*, “Language models of protein sequences at the scale of evolution enable accurate structure prediction,” *bioRxiv*, 2022. DOI: [10.1101/2022.07.20.500902](https://doi.org/10.1101/2022.07.20.500902).
- [6] R. Bommasani, D. A. Hudson, E. Adeli, *et al.*, “On the opportunities and risks of foundation models,” 2021. arXiv: [arXiv:2108.07258](https://arxiv.org/abs/2108.07258).
- [7] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “Bert: Pre-training of deep bidirectional transformers for language understanding,” 2019.
- [8] T. B. Brown, B. Mann, N. Ryder, *et al.*, “Language models are few-shot learners,” 2020.
- [9] E. Nguyen, M. Poli, M. G. Durrant, *et al.*, “Sequence modeling and design from molecular to genome scale with evo,” *bioRxiv*, 2024. DOI: [10.1101/2024.02.27.582234](https://doi.org/10.1101/2024.02.27.582234).
- [10] M. Hao, J. Gong, X. Zeng, C. Liu, Y. Guo, X. Cheng, T. Wang, J. Ma, L. Song, and X. Zhang, “Large scale foundation model on single-cell transcriptomics,” *bioRxiv*, 2023. DOI: [10.1101/2023.05.29.542705](https://doi.org/10.1101/2023.05.29.542705).
- [11] F. Yang, W. Wang, F. Wang, Y. Fang, D. Tang, J. Huang, H. Lu, and J. Yao, “Scbert as a large-scale pretrained deep language model for cell type annotation of single-cell rna-seq data,” *Nat. Mach. Intell.*, vol. 4, no. 10, pp. 852–866, Sep. 2022.
- [12] H. Cui, C. Wang, H. Maan, K. Pang, F. Luo, N. Duan, and B. Wang, “Scgpt: Toward building a foundation model for single-cell multi-omics using generative ai,” *Nat. Methods*, Feb. 2024.
- [13] Z. Zhang, M. Xu, A. R. Jamasb, V. Chenthamarakshan, A. Lozano, P. Das, and J. Tang, “Protein representation learning by geometric structure pretraining,” in *The Eleventh International Conference on Learning Representations*, 2023.

- [14] L. Wang, H. Liu, Y. Liu, J. Kurtin, and S. Ji, “Learning hierarchical protein representations via complete 3d graph networks,” in *The Eleventh International Conference on Learning Representations*, 2023.
- [15] E. C. Alley, G. Khimulya, S. Biswas, M. AlQuraishi, and G. M. Church, “Unified rational protein engineering with sequence-based deep representation learning,” *Nat. Methods*, vol. 16, no. 12, pp. 1315–1322, Dec. 2019.
- [16] M. Heinzinger, A. Elnaggar, Y. Wang, C. Dallago, D. Nechaev, F. Matthes, and B. Rost, “Modeling aspects of the language of life through transfer-learning protein sequences,” *BMC Bioinformatics*, vol. 20, no. 1, p. 723, Dec. 2019.
- [17] G. Derevyanko, S. Grudinin, Y. Bengio, and G. Lamoureux, “Deep convolutional networks for quality assessment of protein folds,” *Bioinformatics*, vol. 34, no. 23, pp. 4046–4053, Jun. 2018. DOI: [10.1093/bioinformatics/bty494](https://doi.org/10.1093/bioinformatics/bty494).
- [18] F. Baldassarre, D. Menéndez Hurtado, A. Elofsson, and H. Azizpour, “GraphQA: protein model quality assessment using graph convolutional networks,” *Bioinformatics*, vol. 37, no. 3, pp. 360–366, Aug. 2020. DOI: [10.1093/bioinformatics/btaa714](https://doi.org/10.1093/bioinformatics/btaa714).
- [19] P. Hermosilla and T. Ropinski, “Contrastive representation learning for 3d protein structures,” 2022.
- [20] P. Chandak, K. Huang, and M. Zitnik, “Building a knowledge graph to enable precision medicine,” *Scientific Data*, vol. 10, no. 1, p. 67, Feb. 2023, ISSN: 2052-4463. DOI: [10.1038/s41597-023-01960-3](https://doi.org/10.1038/s41597-023-01960-3).
- [21] M. Rotmensch, Y. Halpern, A. Tlimat, S. Horng, and D. Sontag, “Learning a health knowledge graph from electronic medical records,” *Scientific Reports*, vol. 7, no. 1, p. 5994, Jul. 2017. DOI: [10.1038/s41598-017-05778-z](https://doi.org/10.1038/s41598-017-05778-z).
- [22] D. Szklarczyk, J. H. Morris, H. Cook, *et al.*, “The STRING database in 2017: Quality-controlled protein–protein association networks, made broadly accessible,” *Nucleic Acids Res.*, vol. 45, no. D1, pp. D362–D368, Jan. 2017.
- [23] K. Huang, P. Chandak, Q. Wang, S. Havaladar, A. Vaid, J. Leskovec, G. Nadkarni, B. S. Glicksberg, N. Gehlenborg, and M. Zitnik, “Zero-shot prediction of therapeutic use with geometric deep learning and clinician centered design,” *medRxiv*, 2023. DOI: [10.1101/2023.03.19.23287458](https://doi.org/10.1101/2023.03.19.23287458).
- [24] S. Pan, L. Luo, Y. Wang, C. Chen, J. Wang, and X. Wu, “Unifying large language models and knowledge graphs: A roadmap,” 2023.
- [25] Y. Li, C. Gu, T. Dullien, O. Vinyals, and P. Kohli, “Graph matching networks for learning the similarity of graph structured objects,” 2019.
- [26] A. Bordes, N. Usunier, A. Garcia-Duran, J. Weston, and O. Yakhnenko, “Translating embeddings for modeling multi-relational data,” in *Advances in Neural Information Processing Systems*, C. Burges, L. Bottou, M. Welling, Z. Ghahramani, and K. Weinberger, Eds., vol. 26, Curran Associates, Inc., 2013.
- [27] D. Maglott, J. Ostell, K. Pruitt, and T. Tatusova, “Entrez gene: Gene-centered information at ncbi,” *Nucleic Acids Res.*, Jan. 2011.

- [28] Gene Ontology Consortium, “The gene ontology resource: Enriching a GOld mine,” *Nucleic Acids Res.*, vol. 49, no. D1, pp. D325–D334, Jan. 2021.
- [29] M. Schlichtkrull, T. N. Kipf, P. Bloem, R. van den Berg, I. Titov, and M. Welling, “Modeling relational data with graph convolutional networks,” in *The Semantic Web*, Springer International Publishing, 2018, pp. 593–607.
- [30] Z. Zhu, C. Shi, Z. Zhang, *et al.*, “TorchDrug: A powerful and flexible machine learning platform for drug discovery,” 2022.
- [31] A. Tiessen, P. Pérez-Rodríguez, and L. J. Delaye-Arredondo, “Mathematical modeling and comparison of protein size distribution in different plant, animal, fungal and microbial species reveals a negative correlation between protein size and protein number, thus providing insight into the evolution of proteomes,” *BMC Res. Notes*, vol. 5, no. 1, p. 85, Feb. 2012.
- [32] C. Orengo, A. Michie, S. Jones, D. Jones, M. Swindells, and J. Thornton, “Cath – a hierarchic classification of protein domain structures,” *Structure*, vol. 5, no. 8, pp. 1093–1109, 1997. DOI: [https://doi.org/10.1016/S0969-2126\(97\)00260-8](https://doi.org/10.1016/S0969-2126(97)00260-8).
- [33] H. M. Berman, J. Westbrook, Z. Feng, G. Gilliland, T. N. Bhat, H. Weissig, I. N. Shindyalov, and P. E. Bourne, “The protein data bank,” *Nucleic Acids Res.*, vol. 28, no. 1, pp. 235–242, Jan. 2000.

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